



GE Medical Systems

Technical Publication

Direction 2239613-100

Revision 0

**GE Medical Systems
LightSpeed Production Upgrade FMI25256**

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LANGUAGE

WARNING

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- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
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WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

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- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
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ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

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Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

End of Section

Revision History

Revision	Date	Reason for change
0	6/25/99	Initial Release of Publication

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List of Effected Pages

PAGES	REVISION	PAGES	REVISION	PAGES	REVISION
1 through 82	0				

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Preface

Publication Conventions

Purpose: This section means to inform the reader on publication conventions used. So that the reader can identify safety and general material that is considered important by its format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task.

Please become familiar with the conventions used within this publication before proceeding.

Section 1.0 Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that are applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.



CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Crush or pinch points
- Sharp objects



DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION



**WARNING
ROTATING
EQUIPMENT
BARE WIRES**

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.



**NOTICE
Equipment
Damage
Possible**

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0 Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

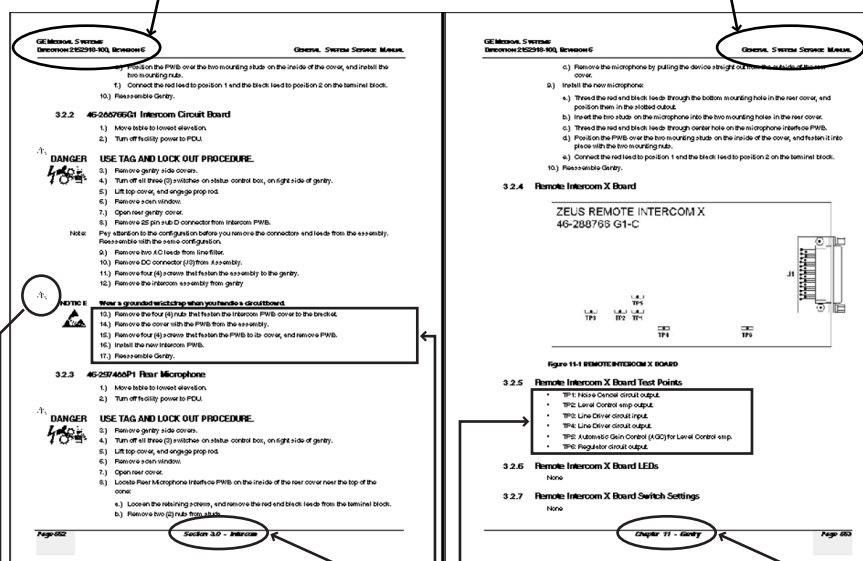
Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents "additional" information that may or may not be relevant.

2.2 Page Layout

Publication Part Number & Revision Number

Publication Title



The current section and its title are always shown in the footer of the left (even) page.

An exclamation point in a triangle is used to indicate important information to the user.

Paragraphs preceded by **Alphanumeric** (e.g. numbers) characters is information that must be followed in a **specific order**.

The current chapter and its title are always shown in the footer of the right (odd) page.

Paragraphs preceded by **symbols** is (e.g. bullets) is information that has **no specific order**.

Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicated directionality. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: This paragraph denotes computer screen fixed output. It's output
Fixed Output is fixed from the sense that it does not vary from application to application. It's the most commonly used style used to indicate filenames, paths and text.

Example: *This paragraph denotes computer screen output that is variable. Its
Variable Output output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>*

Example: **This paragraph denotes fixed input. It's typed input that will
Fixed Input not vary from application to application. Fixed text the user is required to supply as input.**

Example: ***This paragraph denotes computer input that can vary from ap-
Variable Input plication to application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.***

End of Section

CHAPTER 1

OVERVIEW

Section 1.0 PURPOSE

The purpose of the FMI is to update the current installed base of LightSpeed QX/i systems with the production release of operating software and hardware. In addition, the customer and Field Engineer will receive updated user and service information.

Section 2.0 EFFECTIVITY

All LightSpeed QX/i systems manufactured before 5-10-99. The gantry model number for these units is 2180482.

Section 3.0 MATERIALS

FMI Kit Part Number 2239511 consists of the following parts:

Part Number	Description	Quantity
2191432	DASM Cable	1
2174128	RJ45 Cable	2
2148905	Smart Prep Option	1
2181101-5	4.5 GB O2 Disk Drive	1
2199806-5	2 x 128MB DIMM Set	2
2169940-12	9GB Octane Disk Drive	1
2142726	MOD Disk	1
2211214-100	Technical Reference Manual	1
2211217-100	Installation Manual General Class	1
2211218-100	Installation Manual Proprietary Class	1
2211220-100	Load From Cold	1
2211221-100	System Service Manual Gen. Class	1
2211222-100	System Service Manual Prop. Class	1
2211223-100	Tips and WorkArounds	1
2211225-100	FRU Manual	1
2211227-100	Schematics Manual	1
2211228-100	Functional Interconnects Manual	1
2240212	M4 Apps Software	1
2240213	M4 OS SW CDROM A	1
2240215	M4 OS SW CDROM B	1
2240216	M4 OS SW CDROM Upgrade	1
2226999	M4 Sherlock CBT	1

Table 1-1 Materials Required

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Chapter 2

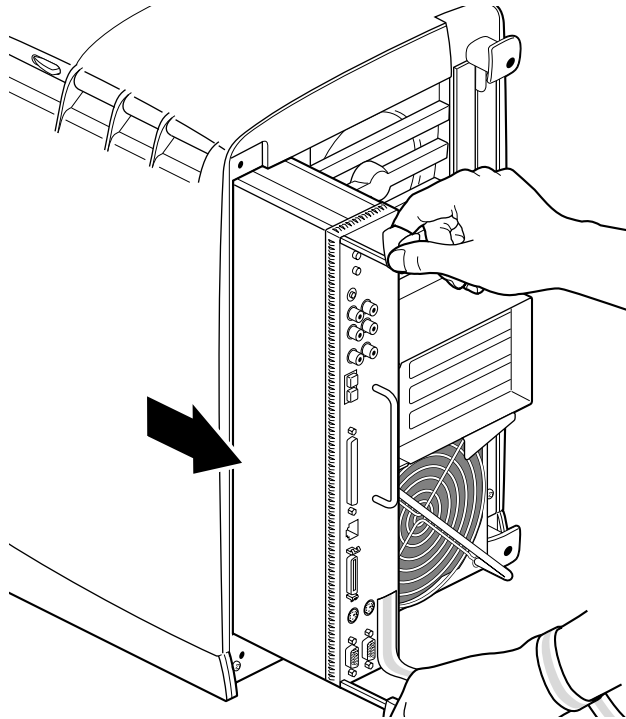
CONSOLE HW UPGRADE

Section 1.0 Record System Information for Upgrade

- 1.) Reference the Load From Cold Manual provided with the FMI kit, Direction 2211220-100 revision 01, to perform to record system information in preparation to upgrade the software.
 - a.) Save System State. Refer to Load From Cold Manual Chapter 2, Section 2.2.
 - b.) Record Key System Information. Refer to Load From Cold Manual Chapter 2, Section 2.3.
 - c.) Perform a System Shut down with pink shutdown button. Refer to Load From Cold Manual Chapter 2, Section 2.4.
 - d.) If you have a keyboard other than English, configure the system for the appropriate keyboard language. Refer to Load From Cold Manual Chapter 2, Section 2.6. If you have an English Keyboard, continue with the next step.

Section 2.0 Installing Extra Memory

2.1 Remove the System Module from the Octane



Look for connector caps to put on the systems module when you remove it. If you do not have any, be very careful not to touch the high density connectors.

Figure 2-1 Removing the System Module

- 1.) Shutdown system and remove power if it has not been done already.
- 2.) Remove the console's front cover. Pull out platform upon which the computer rests. Release its tie down strap if present. The tie down strap should not be present unless the console is in shipment.
- 3.) Remove the locking bar (if applicable). Disconnect all cables from the rear of the Octane. Pull out Octane and work on it away from the console if desired.



NOTICE



Wear a grounded ESD wrist strap. Place removed electronic parts on or in an antistatic surface.



CAUTION



Wait five minutes after power is off before you continue. Let it cool.

- 4.) Loosen the captive screw in the sliding handles on the top and bottom.
- 5.) Pull both handles at the same time until they are fully extended.
- 6.) Grasp the module handle with your left hand and place your right hand against the top of the computer's back. Pull the module out without allowing the delicate connectors on its back edge to touch anything.
- 7.) Place the system module on an antistatic surface with the component side up.
- 8.) Place a cap on each compression connector. If your system does not have one, be very careful not to touch the high density connectors.

HOST (OCTANE) DIMM MEMORY

The NEW (Octane) host configuration uses *512 MB of CPU memory*. You will be removing the existing DIMMS and replacing them with the new ones provided, unless your system has a memory option like 3D. The optional memory will remain in system. DIMM sockets S1 and S2 each will contain a 128MB module for a 256 MB bank. Sockets S3 and S4 each contain a 128MB DIMM for a 256MB bank. This yields the 512MB. The existing DIMMS are 64MB which gave 256MB total.

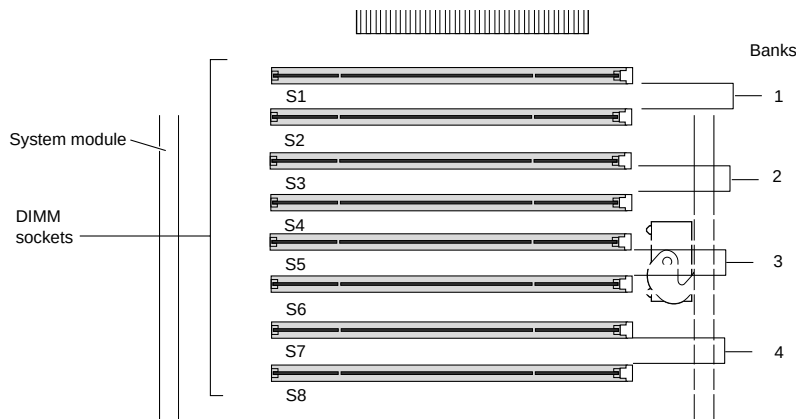


Figure 2-2 (Octane) Host - DIMM Socket and Bank Identification

There are eight DIMM sockets and four banks. A bank must be completely filled with identical modules or empty. Bank 1 (sockets 1 and 2) must always be filled. Banks are filled sequentially; when Bank 1 is full, fill Bank 2, Bank 3 and then Bank 4. This means you must not skip a bank. See [Figure 2-2](#).



NOTICE



**Avoid
Touching
Bristles**

- **Memory modules are extremely sensitive to static electricity. Use an ESD wrist strap and handle with care. Also be aware that the heat sinks inside the computer become very HOT.**
- **The DIMMs are located near a delicate compression connector. Be extremely careful not to touch the compression connector gold bristles.**

2.2 Replacing DIMMs

- 9.) Remove DIMMS from slots S1, S2, S3, and S4. Replace with new DIMMS provided as described in the following steps. Keep Optional Memory for things like 3D in the remaining slots.



NOTICE



Do not touch the connector near the DIMM removal levers.

- 10.) As shown in [Figure 2-3](#), press down on the latch at (A), near the end of the DIMM socket. The DIMM partially ejects from the socket. It can then be removed as shown in (B).

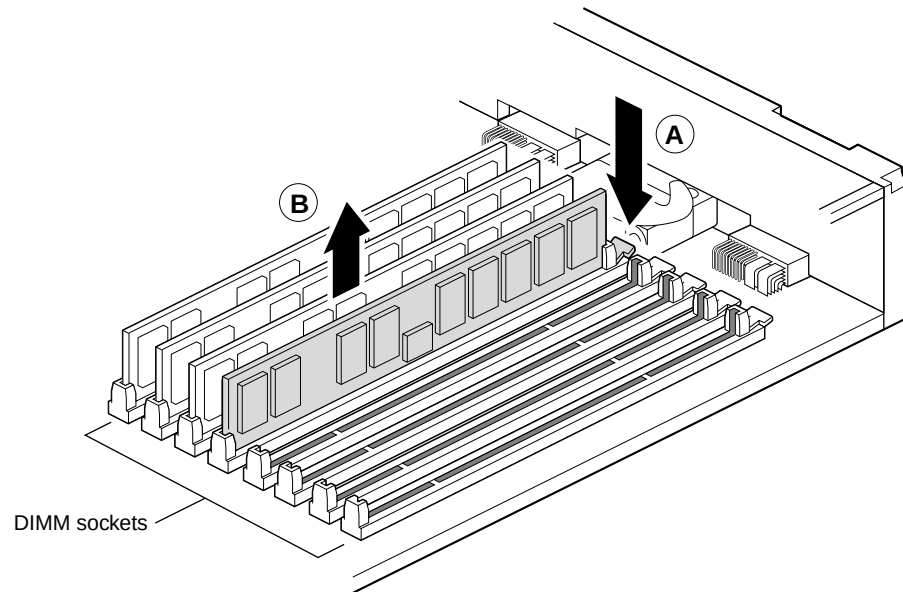


Figure 2-3 Host (Octane) DIMM Removal

2.3 Installing DIMMS

- 1.) Insert the DIMM into the socket, gently but firmly. You can hear a click as it is seated, and the latch on the end of the socket will move up. DIMMs are notched on the bottom so that they cannot be inserted incorrectly. See [Figure 2-4](#).

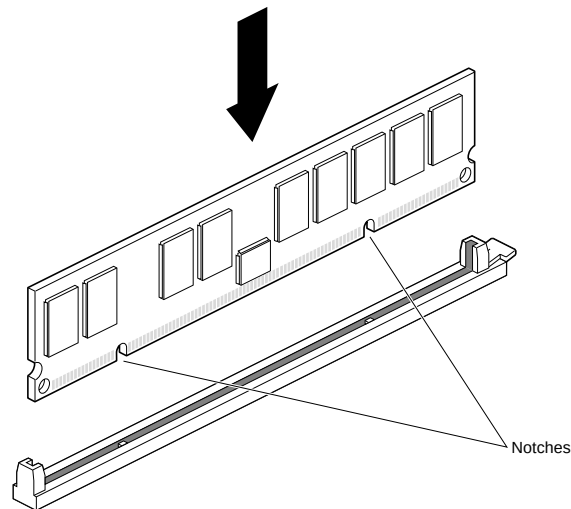


Figure 2-4 (Octane) Host DIMM Installation

- 2.) Check to be sure both sockets in the bank are full. DIMMs must be installed in pairs. **If your system had a memory option for things like 3D, make sure the optional memory is installed in the next available slots.** Remember to remove the C-pop covers on the system module. You have finished installing memory and are ready to replace the system module.
- 3.) Replace the system module in the Octane.

COMMON MISTAKES:

- DIMM Sockets not populated Correctly - Both sockets in a DIMM bank must be either empty or populated. If you are removing one DIMM and not replacing it immediately, also remove the other DIMM in the bank and replace it when you install a new DIMM.
- DIMM not seated properly - Before replacing a memory module, check that all are seated correctly in their slots, the first bank has two DIMMs that are exactly the same. The second bank, if used, has two DIMMs that are exactly the same.
- Incorrect memory combinations - Memory is installed correctly when it is vertical and perpendicular to the motherboard, and the latches on the both sides fit snugly around it. If the memory module appears to be leaning, wear an ESD wrist strap and push it into a vertical position.

Section 3.0 Console RJ45 Cable Replacement

- 1.) Shutdown Applications and power off console.
- 2.) Replace the RJ45 Cables with the new ones provided in the FMI kit. Refer to [Figure 2-5](#). The RJ45 cable is the one mark item 25. There is one for the Octane and one for the O2 computer.
- 3.) Reconnect the cables at the LAN switch: Octane cable in the bottom connection, and the O2 cable in the next to bottom connection.

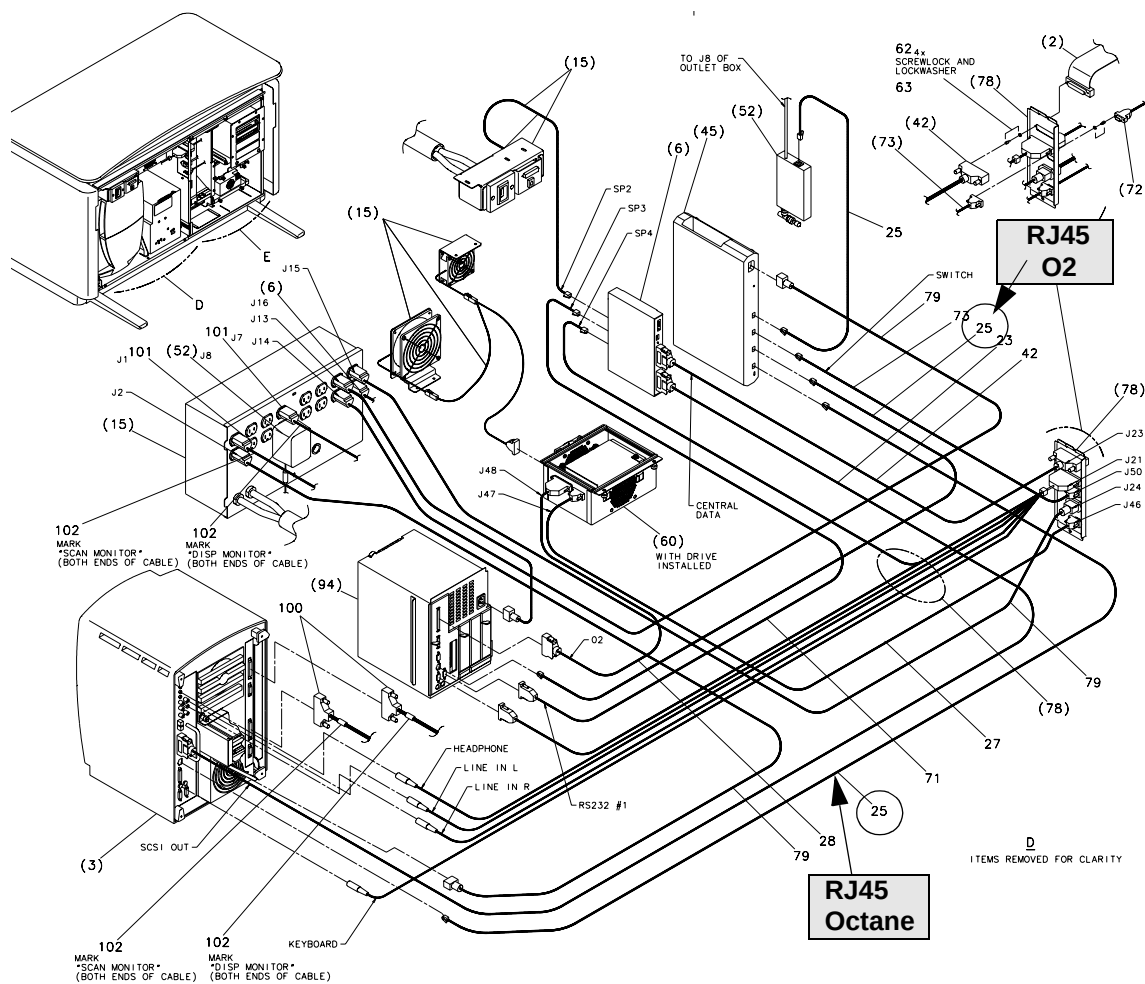


Figure 2-5 Console Assembly (2180551, Sheet 3, Rev)

Section 4.0 DASM Cable Replacement

Note: The DASM does NOT fit in the console. Place the DASM in the space beneath the console tabletop. Many systems were shipped with short DASM cables that would not allow it to be positioned in the proper place. The FMI kit contains a longer cable.

- 1.) Refer to [Figure 2-7](#).
- 2.) Remove the old DASM/SCSI cable from the Octane to the DASM. This is cable A.
- 3.) Replace the old DASM Cable with the new longer one. The new cable will allow the DASM to be placed under the console table top. Route cables and place DASM under console table top.

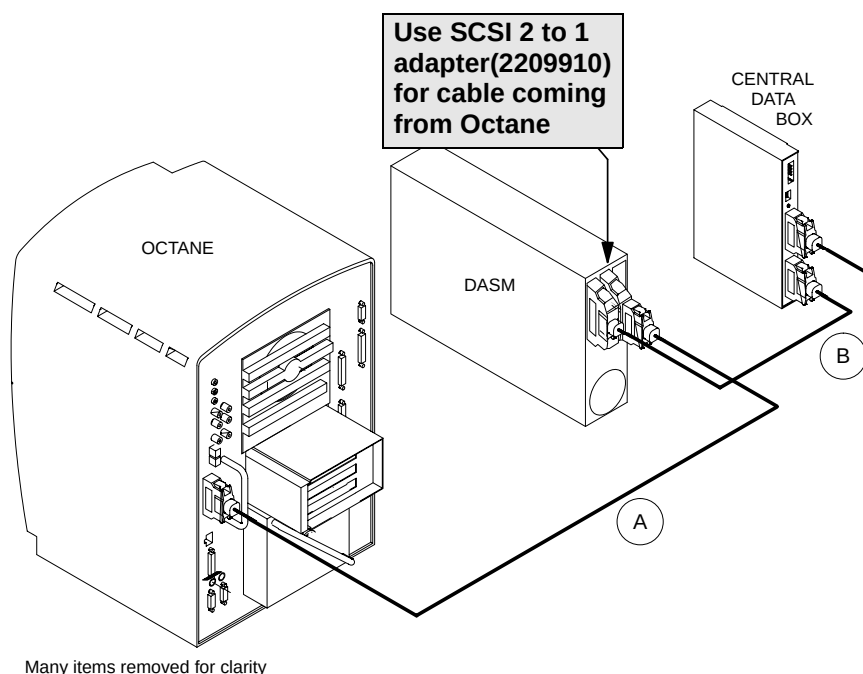


Figure 2-6 SCSI Cable Configuration with DASM

- 4.) Make sure there is enough slack in cables to pull out tray in console.
- 5.) Plug DASM power cord into J10, [Figure 2-7](#). Attach remaining cable(s) from camera to DASM.

J#	Device	J#	Device
J1	Rx Monitor	J9	Vieb
J2	Powered Transceiver	J10	DASM
J3	Kieb	J11	Image Printer
J4	VCR	J12	Modem
J5	Printer	J13	Remote Monitor
J6	Video Splitter	J14	Spare
J7	Display Monitor	J15	Central Data Box
J8	Octane	J16	EMC Chassis

Table 2-1 Power Panel Outlet Assignments

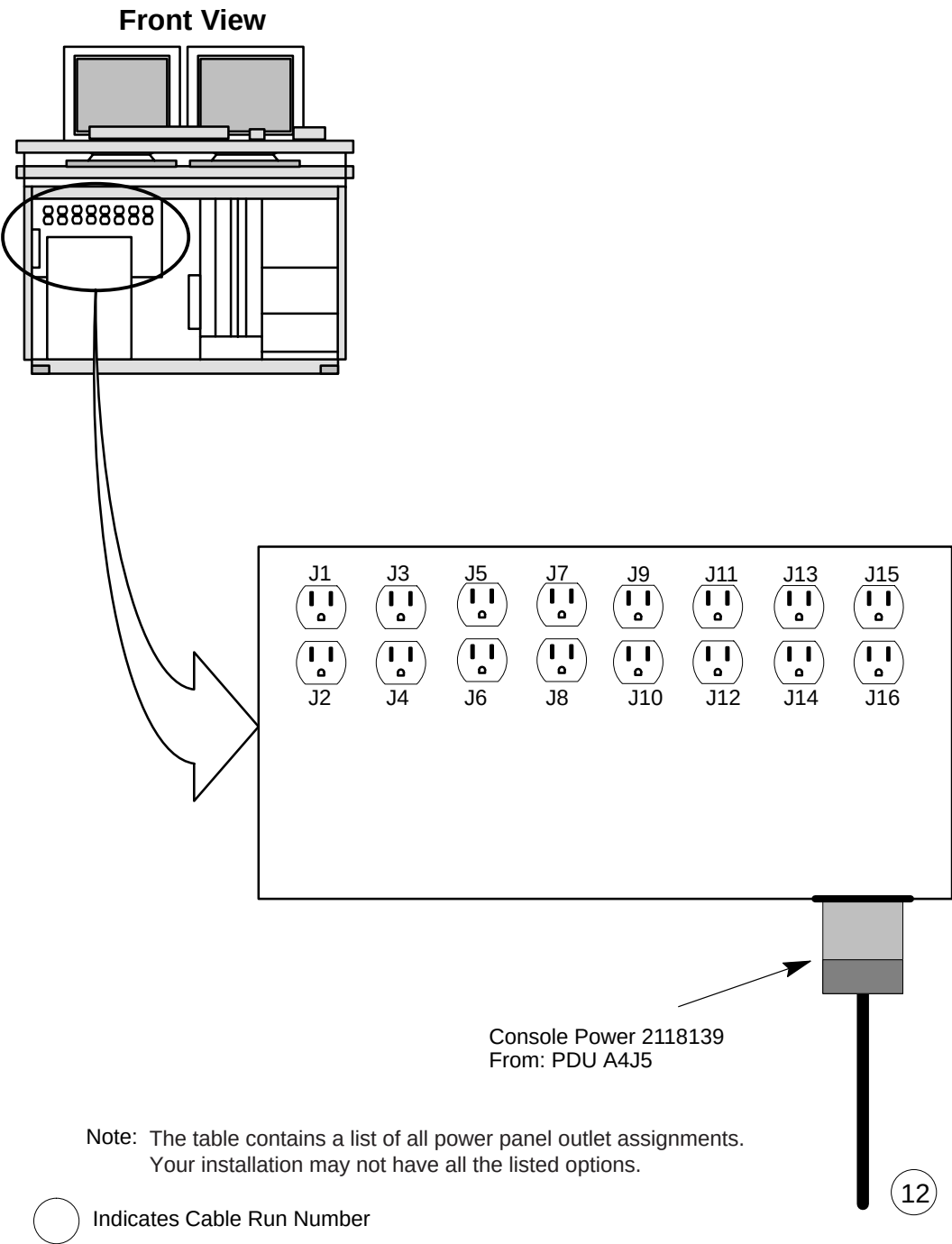


Figure 2-7 Console Power Panel

Section 5.0 Host (Octane) Hard Drive

The Octane hard drive is supplied by SGI. *Place the primary system drive in the bottom bay on the front of the computer.* If a second system disk is added, place it in the middle bay. The host assigns their SCSI IDs. The primary disk is SCSI ID 1 (dks0d1sN). The one inserted in the middle becomes SCSI ID 2 (dks0d2sN).

Note: The present configuration has a 4.5GB drive in bay1, and a 9GB drive in bay 2. The new configuration after this FMI is 2 9GB drives in bay 1 and 2.

HOW TO REPLACE OR INSTALL AN OCTANE SYSTEM DRIVE

- 1.) Shutdown system and remove power.
- 2.) Press both bezel release buttons on front upper sides.
- 3.) Tilt the cover forward and lift to remove it. Lift the handle of the top drive so it is not hit when removing the bottom drive.

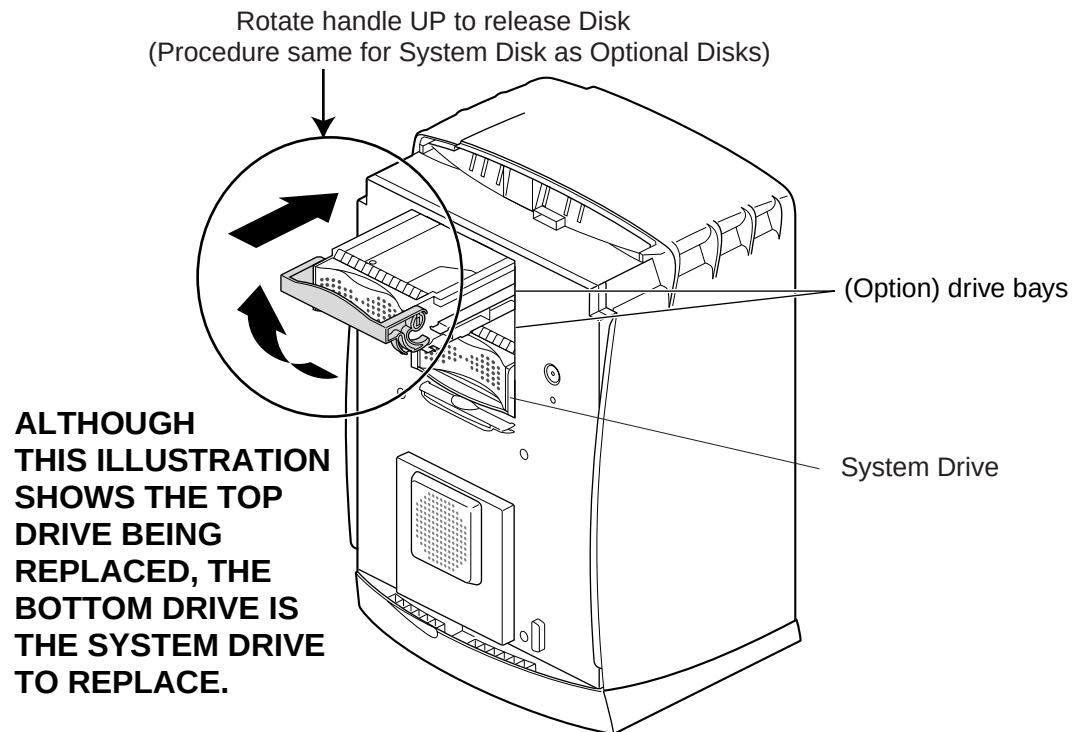


Figure 2-8 Host (Octane) System Drive

- 4.) On the bottom drive, the bay 1 drive, lift the drive's handle to the horizontal position and *gently* slide it into the bay. *Pushing a drive in with too much force can damage it. Seat the drive carefully but firmly.*
- 5.) When it is flush with the chassis, rotate the handle down to lock it.
- 6.) The drive in the top bay will remain. It will not be changed.
- 7.) After the system is powered on, the disks will need to be configured. For this FMI, the disks are coming with software pre-loaded. A Load From Cold is not necessary. Do not re-install the Octane. That will be covered later in the procedure.

Section 6.0 Replacing the O2/SBC System Drive

Before replacing the system drive, you must turn off the workstation, remove the power cable and other exterior cables, and attach the wrist strap. Follow these steps:

- 1.) Turn off the workstation.
- 2.) Some sites may feel the system drive can be removed without pulling out the O2 computer. If you feel you can do this, skip to the next step. Otherwise, remove the power cable and other cables from the rear. Remove the O2 from the console if desired.
- 3.) Pull down the lever on the left side of the system drive module to release it ([Figure 2-9 on page 32](#)).



Figure 2-9 Releasing the Drive Tray

- 4.) Pull the system drive module out of the chassis using the small handle.
- 5.) Slide the replacement system drive module into the system drive bay ([Figure 2-10 on page 32](#)).

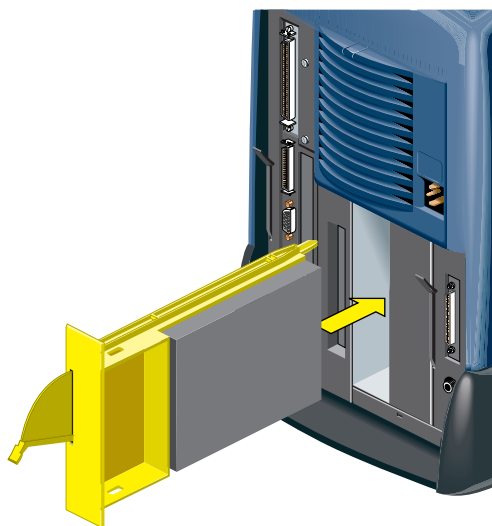


Figure 2-10 Installing the Replacement System Drive Module

- 6.) Push the lever up to lock the tray in place ([Figure 2-11 on page 33](#)).

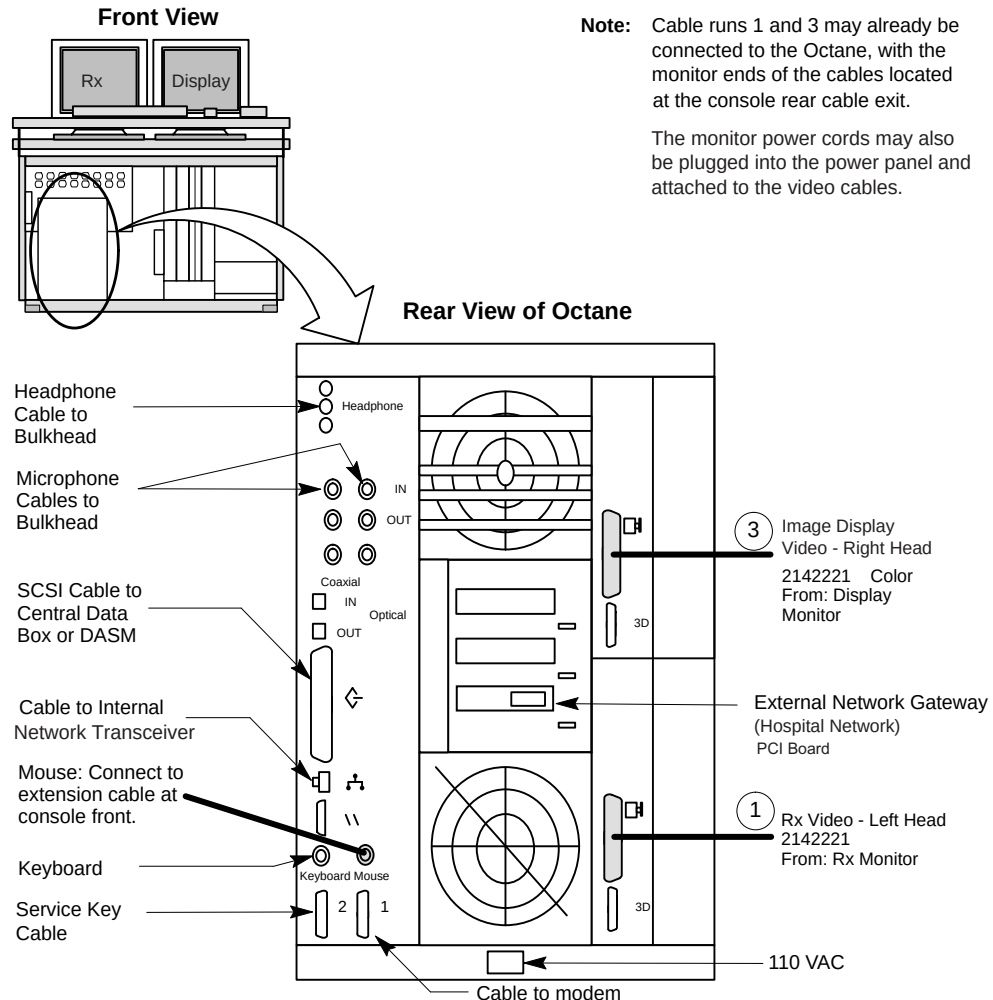


Figure 2-11 Securing the System Drive Module

- 7.) For this FMI, the drive is pre-loaded with software. A Load From Cold will not be necessary. Go to the next section to continue the upgrade.

Section 7.0 Connect all cables, return console to operating configuration, and power on console.

- 1.) Return the Octane and O2 to the console if they were removed.
- 2.) Connect all cables in the console. Refer to [Figure 2-12](#) and [Figure 2-13](#) below.
- 3.) Prepare to power on the console.



Note: Remember to leave sufficient cable length for a service loop. Make sure you can fully extend the service tray, and return it to the storage position without pulling or pinching the cables.

○ Indicates Cable Run Number

Figure 2-12 Rear View of Octane Computer Carrier

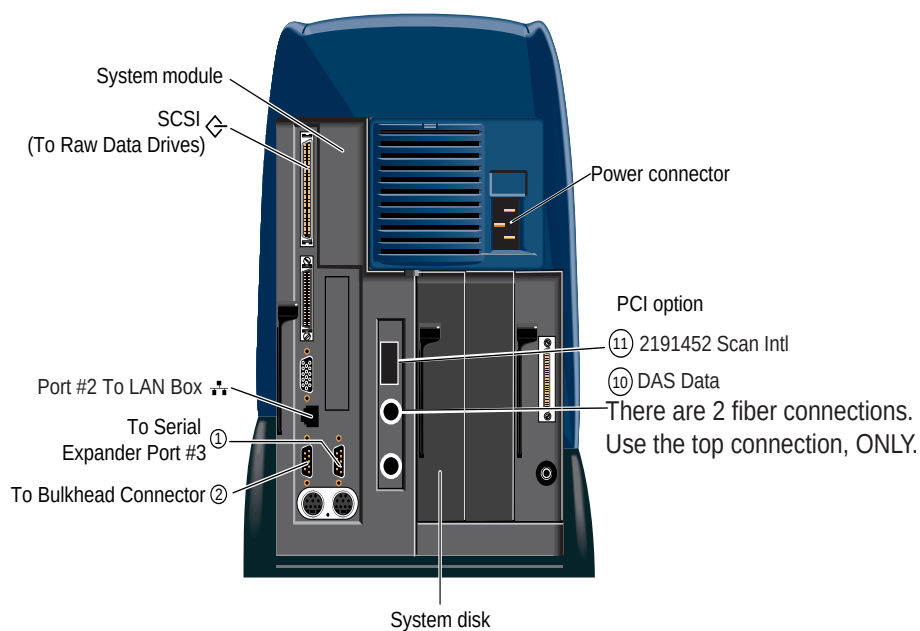


Figure 2-13 Rear View of O2 Computer

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Chapter 3

Software Upgrade and Configuration

Section 1.0 Configure new drive with the Software

- 1.) Power on the System. The system will attempt to boot up, but will fail to go to applications level. Do not allow the system to come up to applications. Press OK to cancel applications start up. If you get a message stating the following: Unable to initialize scan data disk handling. Size of scan data disks appear incorrect. This can occur if there is no power to the scan data disks. (Error number 230011004). You need to go to step 2 and reset the O2 computer.

Note: Ignore any messages: Some file systems failed to mount; may be unable to reach multi-user state

- 2.) Press the reset button on the O2 computer. The reset button is the button to the left of the green/red LED on the front of the O2 computer.

Note: The SBC must be up and running UNIX before proceeding. The state of the SBC can be verified by opening a unix shell on the OC and typing `cu sbc` and pressing ENTER. If this comes back with a login prompt, UNIX is up. Type `~.` (tilde dot) to cancel the SBC `cu` session.

- 3.) Open a Unix shell. A UNIX shell can be opened from the TOOLCHEST menu.

- 4.) Log in as root. Type: `su -` and ENTER. Root password is **#bigguy**.

- 5.) Type: **install2Disk**

After the script completes verify the following lines appear:

The chunk specified already exists

Finished with the disk configuration

- 6.) Insert the System State MOD into the MOD drive

- 7.) Type: **mountMOD**

- 8.) Type: **cp /MOD/service_mod_data/system_state/INFO /usr/g/config/INFO**

- 9.) Type: **unmountMOD**

- 10.) Type: **reconfig**

- 11.) When the LightSpeed Install window appears, click on CONFIG.

- 12.) Run a reconfig of the system. Select YES on the System Settings GUI for Regenerate Database. Refer to **Load From Cold Manual**, Chapter 2, Section 2.10, Steps 5-9.

- 13.) When the re-configuration is complete, the reboot prompt window appears. Open a Unix shell.

- 14.) Login in as root, type: **(su -)** and press ENTER. Root Password is **# bigguy**.

a.) In the new shell window type: **rsh sbc**

b.) In the new shell window type: **rm -f /usr/g/state/tubeCooling.stf**

c.) Click on YES in the reboot prompt window to reboot the system

d.) The system will automatically login as `ctuser` after the reboot. Click on OK on the Autostrada Disabled popup message

- 15.) Perform the following steps from the **Load From Cold Manual, 2211220-100**.
- a.) Set the date/time on OC and SBC and set security. Refer to Load From Cold Manual, Chapter 2, Section 2.11. **Setting security is a necessary action to prevent external access to the system. When you are asked if you wish to overwrite the existing file, answer yes.**
 - b.) Start up CT applications. Refer to Load From Cold Manual, Chapter 2, Section 2.12.
 - c.) Restore System State. Refer to Load From Cold Manual, Chapter 2, Section 2.13.
 - d.) Download Flash Firmware. Refer to Load From Cold Manual, Chapter 2, Section 2.14.
 - e.) Add Software Options. Refer to Load From Cold Manual, Chapter 2, Section 2.15.
 - f.) Install Cameras if needed. Refer to Load From Cold Manual, Chapter 2, Section 2.16.
 - g.) Shutdown System. Refer to Load From Cold Manual, Chapter 2, Section 2.17.
 - h.) Verify System Operation. Refer to Load From Cold Manual, Chapter 2, Section 2.18.

Section 2.0 Network Set Up with new Software

The following describes the different GUI's that will be used when configuring the networks with the new release software. There are three configurations described, LightSpeed, CT/i, and AWW. This release of software has new network settings, and you may be unfamiliar with the settings.

LIGHTSPEED OR 3.X AWW WORKSTATION NETWORK SETTINGS

Note: Remember that all machines networked to the LightSpeed system need to have the Host name and possible AE Title corrected to function properly

Figure 2-1 LightSpeed Network Settings

Host Name - restored from State MOD, but it had a _oc on the end of the name which is not required now.

Network Address - Restored from State MOD, and should be available from Sys Admin of site.

Network Protocol - LightSpeed systems support full DICOM capability. Make sure the equipment you are connecting to supports DICOM, or the functionality will be reduced.

Port Number - Selected automatically upon choosing Network Protocol.

AE Title - This is the same as the Host name. Be careful that the _oc is removed from these names. Restored from State MOD.

Comments - Optional

Items below Comments are new fields:

Archive Node - If the machine is used as an archive node for the CT department, this should be activated as yes.

Access to Local Host - These settings refer to this system, not others. You must select if the user wishes to send images to

this equipment and have query/retrieve available from this equipment. Most cases will be Yes.

Custom Search - Allows user to selectively choose parameters to narrow searches to if they use a DICOM archive system. Most cases will be ON.

HOST CONFIG FILE EXAMPLE - EXTRACT FROM A /ETC/HOSTS FILE

```
...
224.0.2.2      sun-rpc.mcast.net
3.231.48.22    msecqx12 This is the Host Name and AE Title
192.9.220.10   ct02_oc ct02_oc0 CT02_OC0 oc OC This is the Suite Name;E aliases
192.9.220.1    ct02_sbc ct02_sbc0 CT02_SBC0 sbc SBC
192.9.220.11   ct02_ice ice ICE
192.9.220.1    sbcorig
```

ADVANTAGE NET SET UP - CT/I OR OLDER AWW STATIONS

The following GUI shows the Network Settings that might be used for an Advantage Net System. Advantage Net Protocol does not offer full compatibility with LightSpeed DICOM formats. This screen looks like a CT/i or AWW early release.

Note: Remember that all machines networked to the LightSpeed system need to have the Host name and possible AE Title corrected to function properly

Figure 2-2 Advantage Net Network Protocol

Host Name - restored from State MOD, but it had a _oc on the end of the name which is not required now.

Network Address - Restored from State MOD, and should be available from Sys Admin of site.

Network Protocol - LightSpeed systems support full DICOM capability. Make sure the equipment you are connecting to supports DICOM, or the functionality will be reduced.

Port Number - Selected automatically upon choosing Network Protocol.

AE Title - This is the same as the Host name. Be careful that the _oc is removed from these names. Restored from State MOD.

Comments - Optional

Items below Comments are new fields:

Archive Node - If the machine is used as an archive node for the CT department, this should be activated as yes. This is unavailable for Advantage Net protocol.

Access to Local Host - These settings refer to this equipment presently being serviced. You must select if the user

wishes to send images to the equipment and have query/retrieve available. Most cases will be Yes. This is unavailable for Advantage Net protocol.

Custom Search - Allows user to selectively choose parameters to narrow searches to if they use a DICOM archive system. Most cases will be ON. This is unavailable for Advantage Net protocol.

Figure 2-3 Custom Search GUI - Search Fields

For more information about Network compatibility, refer to the LightSpeed Technical Reference Manual. There is a full explanation of the fields and the network performance of different scanners and workstations. The manual number is 2211214-100.

Section 3.0 SdCRHosts file information

On the host, there is a file that contains information relating to the Network set up. The file can be found by looking at: /usr/g/ctuser/Prefs/SdCRHosts

TYPICAL OUT OUT WOULD APPEAR AS THE FOLLOWING:

3.7.52.135	engbay04	# 2	4006	engbay04	No comment	0	1	1	0
3.7.52.151	engbay26	# 2	4006	engbay26	No comment	0	1	1	0
3.7.52.12	engbay16	# 1	4005	engbay16	No comment	0	1	0	1
A	B	C	D	E	F	G	H	I	J

The fields in the above file relate to the Network GUI settings.

A - IP Address

B - Hostname

C - # is a space, 1 is Advantage Net Protocol, 2 is DICOM Protocol

D - Port Number

E - AE Title

F - Comment Fields

G - Archive Node - 1 = Yes, 0 = No

H - Send Images - 1 = Yes, 0 = No

I - Query/Retrieve - 1 = Yes, 0 = No

J - Custom Search - 1 = Yes, 0 = No

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Chapter 4

Enter the Detector ID (Serial Number).

- 1.) Select SYSTEM INTEGRATION
- 2.) Select COMPUTER INTERGATION
- 3.) Select START
- 4.) Select ENTER DETECTOR ID
- 5.) Open gantry and look for rating plate/bar code located on the end of the detector. Then enter the serial number of the detector. Continue with the procedure.

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Chapter 5

Perform a Full Detailed Calibration

Note: The full detailed calibration should include Z-Slope Cal, Phantom Cal, and Water Cal (CT# Adjust). Auto CT# adjust is having some problems. It may be necessary to hit resume several times during the process. If Auto CT# adjust does not work, perform CT# adjustments with Manual CT # Adjust. **The Center Phantom process has a time limit programmed in the software. If you take longer than 10 minutes to center the phantom, the calibration will be terminated. You will have to start calibration again to continue.**

Section 1.0 Calibration Process B

Note: Take care to align the water filled area of the phantom in the SFOV during water cals. Locate the black positioning markers over the resolution portion of the phantom. Scan the section of the phantom that starts about 60mm (2 3/8 inches) in front of the black positioning markers. Follow this calibration process when the system arrives from the factory without a set of calibration files.

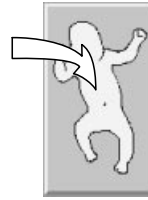
- 1.) Perform Z-Slope Cal by selecting the function from Scanner Utilities or the Service Desktop. It runs air scans. Make sure x-ray beam is clear.
- 2.) Select SCANNER UTILITIES, DETAILED CALIBRATION, and then choose the user select option.
- 3.) Use a bubble level and the Z-axis knob on the phantom holder to level the phantom.
- 4.) Select the following parameters from the Air Calibration screen.
 - a.) Calibration Type: Small/Head, Large/Body
 - b.) KV: Select all kV settings.
 - c.) Aperture: All
 - d.) Focal Spot: All
- 5.) Acquire the Air and Phantom Calibrations. When the scans complete, windows open with detailed instructions to center and scan the calibration phantoms.
- 6.) When you finish the calibration process, continue to the Auto CT N Number Adjustment, [Section 3.0](#).

Section 2.0 How to Scan with Protocols

Locate the Manufacturing and Installation protocols under Infant area 20.

Note: Manufacturing and Service share this Protocol list. Different product option offerings also use this list. Carefully follow the scan section instructions, and verify you acquired the images with the correct technique before filling out the data sheet. Otherwise you may troubleshoot an image problem that only exists because you used the wrong technique.

- 1.) Select the NEW PATIENT icon.
- 2.) Enter the Patient ID.



- 3.) Click left on the Infant anatomical figure icon:



- 4.) Click left on the area below the infant's feet (inside circle):
- 5.) Select a protocol from the list, to display the corresponding view edit screen. Optional Method: Enter the Protocol Number into the Protocol Number Field on the Exam Rx Screen.

Section 3.0 AUTO CT N Number Adjustment

Note: If it has been more than 30 minutes since calibration scans, rerun FASTCAL. N Number Adjustment needs the tube to be warm, or CT numbers will change. FASTCAL takes about 15 minutes.

Run the small and large phantom CT N number adjustment before you acquire the Image Series, or any time you doubt the N# values. The calibration procedure modifies the pcal (phantom cal) vector to assign water a CT number equal to zero. After a successful N number adjustment, any water phantom scanned against a small or large CAL has a CT number near zero.

- 1.) Select SCANNER UTILITIES, WATER CALIBRATION
- 2.) Follow directions on the screen.
- 3.) The program will automatically make adjustments
- 4.) If the program is unable to make adjustments, it will terminate at the technique it is unable to adjust.
- 5.) Troubleshoot with manual CT # Adjust before continuing.

Section 4.0 MANUAL CT N Number Adjustment

Note: If Auto CT N number adjustment completed without error, then proceed to N Number Scan Analysis. It is preferred to run the CT N Number service protocol, and evaluate the CT N numbers. this will match the order of the CT N Number table. If you review the **retro** images **only** from the automated process, the images are out of order as described in the table. Both retro images and/or the newly scanned images should meet the 0+/- 1.5 specification. No adjustments are required if specifications are met.

If it has been more than 30 minutes since calibration scans, rerun FASTCAL. N Number Adjustment needs the tube to be warm, or CT numbers will change. FASTCAL takes about 15 minutes.

Run the small and large phantom CT N number adjustment before you acquire the Image Series, or any time you doubt the N# values. The calibration procedure modifies the pcal (phantom cal) vector to assign water a CT number equal to zero. After a successful N number adjustment, any water phantom scanned against a small or large CAL has a CT number near zero.

- 1.) Select EXAM RX
- 2.) Place the QA phantom on the phantom holder. Align the water section with the laser lights.

Note: Take care to align the water filled area of the phantom in the SFOV during water cals. Locate the black positioning markers over the resolution portion of the phantom. Scan the section of the phantom that starts about 60mm (2 3/8 inches) in front of the black positioning markers.

- 3.) Select SCANNER UTILITIES
- 4.) Select CENTER PHANTOM from the Scanner Utilities Menu. Center the Phantom.
- 5.) Select the Service Protocol CT N NUMBERS
- 6.) Set the center of the water section of the QA phantom to S0.0.
- 7.) Use the scan protocols listed in [Table5-1](#) for 80kV, [Table5-2](#) for 100kV, [Table5-3](#) for 120IV, and [Table5-4](#) for 140kV.
- 8.) Complete all scans in the CT N numbers protocol.
- 9.) Proceed to section Section 5.0 on page 49.

#	kV	Aperture	Stop	Start	CAL	F spot	mA
1	80	4x1.25	I1.8	S1.8	Small	Small	200
2	80	4x1.25	I1.8	S1.8	Large	Small	200
3	80	4x2.5	I3.8	S3.8	Small	Small	200
4	80	4x2.5	I3.8	S3.8	Large	Small	200
5	80	4x3.75	I5.6	S5.6	Small	Small	200
6	80	4x3.75	I5.6	S5.6	Large	Small	200
7	80	4x5	I7.5	S7.5	Small	Small	200
8	80	4x5	I7.5	S7.5	Large	Small	200
9	80	4x1.25	I1.8	S1.8	Small	Large	350
10	80	4x1.25	I1.8	S1.8	Large	Large	350
11	80	4x2.5	I3.8	S3.8	Small	Large	350
12	80	4x2.5	I3.8	S3.8	Large	Large	350
13	80	4x3.75	I5.6	S5.6	Small	Large	350
14	80	4x3.75	I5.6	S5.6	Large	Large	350
15	80	4x5	I7.5	S7.5	Small	Large	350
16	80	4x5	I7.5	S7.5	Large	Large	350

Table 5-1 80kV Protocols

#	kV	Aperture	Stop	Start	CAL	F spot	mA
1	100	4x1.25	11.8	S1.8	Small	Small	200
2	100	4x1.25	11.8	S1.8	Large	Small	200
3	100	4x2.5	13.8	S3.8	Small	Small	200
4	100	4x2.5	13.8	S3.8	Large	Small	200
5	100	4x3.75	15.6	S5.6	Small	Small	200
6	100	4x3.75	15.6	S5.6	Large	Small	200
7	100	4x5	17.5	S7.5	Small	Small	200
8	100	4x5	17.5	S7.5	Large	Small	200
9	100	4x1.25	11.8	S1.8	Small	Large	300
10	100	4x1.25	11.8	S1.8	Large	Large	300
11	100	4x2.5	13.8	S3.8	Small	Large	300
12	100	4x2.5	13.8	S3.8	Large	Large	300
13	100	4x3.75	15.6	S5.6	Small	Large	300
14	100	4x3.75	15.6	S5.6	Large	Large	300
15	100	4x5	17.5	S7.5	Small	Large	300
16	100	4x5	17.5	S7.5	Large	Large	300

Table 5-2 100kV Protocols

#	kV	Aperture	Stop	Start	CAL	F spot	mA
1	120	4x1.25	11.8	S1.8	Small	Small	200
2	120	4x1.25	11.8	S1.8	Large	Small	200
3	120	4x2.5	13.8	S3.8	Small	Small	200
4	120	4x2.5	13.8	S3.8	Large	Small	200
5	120	4x3.75	15.6	S5.6	Small	Small	200
6	120	4x3.75	15.6	S5.6	Large	Small	200
7	120	4x5	17.5	S7.5	Small	Small	200
8	120	4x5	17.5	S7.5	Large	Small	200
9	120	4x1.25	11.8	S1.8	Small	Large	250
10	120	4x1.25	11.8	S1.8	Large	Large	250
11	120	4x2.5	13.8	S3.8	Small	Large	250
12	120	4x2.5	13.8	S3.8	Large	Large	250
13	120	4x3.75	15.6	S5.6	Small	Large	250
14	120	4x3.75	15.6	S5.6	Large	Large	250
15	120	4x5	17.5	S7.5	Small	Large	250
16	120	4x5	17.5	S7.5	Large	Large	250

Table 5-3 120kV Protocols

#	kV	Aperture	Stop	Start	CAL	F spot	mA
1	140	4x1.25	I1.8	S1.8	Small	Small	170
2	140	4x1.25	I1.8	S1.8	Large	Small	170
3	140	4x2.5	I3.8	S3.8	Small	Small	170
4	140	4x2.5	I3.8	S3.8	Large	Small	170
5	140	4x3.75	I5.6	S5.6	Small	Small	170
6	140	4x3.75	I5.6	S5.6	Large	Small	170
7	140	4x5	I7.5	S7.5	Small	Small	170
8	140	4x5	I7.5	S7.5	Large	Small	170
9	140	4x1.25	I1.8	S1.8	Small	Large	200
10	140	4x1.25	I1.8	S1.8	Large	Large	200
11	140	4x2.5	I3.8	S3.8	Small	Large	200
12	140	4x2.5	I3.8	S3.8	Large	Large	200
13	140	4x3.75	I5.6	S5.6	Small	Large	200
14	140	4x3.75	I5.6	S5.6	Large	Large	200
15	140	4x5	I7.5	S7.5	Small	Large	200
16	140	4x5	I7.5	S7.5	Large	Large	200

Table 5-4 140kV Protocols

Section 5.0N Number Scan Analysis

- 1.) Select IMAGE WORKS (not Exam Rx) to display the images you acquired for the CT N Number Analysis.
- 2.) Use the List Select function to display the first group of images.
- 3.) Select Review Layouts.
 - a.) Display the images in groups of four, so you can view all detector rows at once for a single scan. In [Section 4.0](#), you acquired 16 scans (64 images) for each kV station.
 - b.) Select a DFOV of 25cm, and use the default ROI that appears on the screen. When displayed in groups of four, by aperture size:
 - * n1 (row 2a) = image in the upper left corner; I image location
 - * n2 (row 1a) = image in the upper right corner
 - * n3 (row 1b) = image in the lower left corner
 - * n4 (row 2b) = image in the lower right corner; S image location
 - c.) Select the box ROI. (The box appears over the center of the image; keep this default position.)
If you type **prop a** in the Speedbar, the ROI propagates to all 4 images.
 - d.) Record the CT numbers of water in the corresponding CT N Number Tables:
 - * 80kV N numbers: [Table 5-5 on page 51](#)
 - * 100kV N numbers: [Table 5-6 on page 51](#)
 - * 120kV N numbers: [Table 5-7 on page 52](#)
 - * 140kV N numbers: [Table 5-8 on page 52](#)

- 4.) If necessary, select SCANNER UTILITIES to open the Utilities window.
- 5.) Select DETAILED CALIBRATION.
- 6.) Select CT NUMBER EDITOR.
- 7.) Select the Browser mode.

- Use the browser to view the old CT numbers for rows 2a, 1a, 1b and 2b.
- Select a button on the GUI to display the corresponding techniques

Note: Browser mode deactivates the APPLY button. Leave the browser mode to modify the row 2a.1a, 1b and 2b values.

- 8.) Select UPDATE mode.

Note: DO NOT engage the Num Lock key on the numeric keypad when you enter the N number values. **Remember** to check the Num Lock key, and *make sure the green light is NOT lit* before you start. You can use the numeric keypad to enter the values, regardless of the state of the Num Lock key, *but* the system refuses to update the N# values with Num Lock engaged. **IF** you enter the values with Num Lock engaged (green light ON) you will have to reenter all 256 N# values.

- 9.) Check the Num Lock key on the numeric keypad. If necessary, press NUM LOCK to turn OFF the green light on the keycap.
- 10.) Select a kV to modify.
- 11.) Type/enter the numbers you recorded in Tables 5-5 through 5-8 into the editor. Use the arrow buttons to cycle through your entries. Check your entries.
- 12.) Finish entering the CT N numbers, then select APPLY. Follow this process to enter all kV CT N number adjustments.

Note: If it has been more than 30 minutes since calibration scans, rerun FASTCAL. N Number Adjustment needs the tube to be warm, or CT numbers will change. FASTCAL takes about 15 minutes.

- 13.) Scan the water phantom, and verify the CT number equals 0 ± 1.5 . If the CT number falls outside the 0 ± 1.5 range, repeat Section 4.0 and 5.0 until the CT number of the water phantom equals 0 ± 1.5 .

RECORD AND ENTER THE DATA FROM LEFT TO RIGHT (TOP TO BOTTOM)

80kV	Small Spot Head (Small) Cal 200mA				Small Spot Body (Large) Cal 200mA			
	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								
80kV	Large Spot Head (Small) Cal 350mA				Large Spot Body (Large) Cal 350mA			
	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								

Table 5-5 N Number Table for 80kV

RECORD AND ENTER THE DATA FROM LEFT TO RIGHT (TOP TO BOTTOM)

100kV	Small Spot Head (Small) Cal 200mA				Small Spot Body (Large) Cal 200mA			
	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								
100kV	Large Spot Head (Small) Cal 300mA				Large Spot Body (Large) Cal 300mA			
	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								

Table 5-6 N Number Table for 100kV

RECORD AND ENTER THE DATA FROM LEFT TO RIGHT (TOP TO BOTTOM)

120kV	Small Spot Head (Small) Cal 200mA				Small Spot Body (Large) Cal 200mA			
	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								
120kV	Large Spot Head (Small) Cal 250mA				Large Spot Body (Large) Cal 250mA			
	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								

Table 5-7 N Number Table for 120kV

RECORD AND ENTER THE DATA FROM LEFT TO RIGHT (TOP TO BOTTOM)

140kV	Small Spot Head (Small) Cal 170mA				Small Spot Body (Large) Cal 170mA			
	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								
140kV	Large Spot Head (Small) Cal 200mA				Large Spot Body (Large) Cal 200mA			
	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)	n1 (row 2a)	n2 (row 1a)	n3 (row 1b)	n4 (row 2b)
4x125								
4x250								
4x375								
4x500								

Table 5-8 N Number Table for 140kV

Section 6.0 Perform Image Quality Checks.

Note: Perform the same image quality checks that are done for a new system installation. Record information on the data sheets in the manual. Return the information to CT Engineering

6.1 Image Series

6.1.1 Scan Protocol

The person who acquires the image series has the responsibility to review the images, and verify they meet the specifications listed on data sheets. Responsibilities also include and means and standard deviation measurements, and keeping a record of failures that occur during the image series.

Unless otherwise stated, use the following scan parameters during the Image Series acquisition:

- Scan FOV equal to display FOV (Field of View)
- 512x512 matrix size

Note: Consider *any* image series scan that does not meet specifications as failing.

For means and standard deviations, 90% of the slices must pass. Any failure on a particular technique requires at least 10 additional slices to evaluate effectively.

Systems with metal-free cradles have a phantom holder with a perpendicular adjustment (Z-axis) knob on it. **Each time you change phantoms**, make sure you use a bubble level, and the Z-axis knob on the phantom holder, to level the phantom.

6.1.2 Data Recording: Means and Standard Deviation

Any failure on a particular technique requires at least a 10 additional slices to evaluate effectively. For means and standard deviations, 90% of the slices must pass.

- Record means to 2 decimal places, and round to the nearest one-tenth, (1decimal place) when you compare the resulting values to the spec.
- Record standard deviations to 2 decimal places, then round off to 1 decimal place, to compare it to the spec.
- Average standard deviations: Use 2 decimal places to average the values, then round off to one place.

Before you record the means and standard deviations, check the image data sheets to determine whether to average the means and standard deviations, or record them slice by slice. Make sure you record all the required Image data on the HHS data sheets.

6.1.3 Term Definitions

Xc

Mean CT number for the specified center coordinates of the phantom image.

AvXc

Average Mean CT number for the center of the phantom image: Average the mean CT value for all specified center coordinates of all slices in an exam.

Xo

Mean CT number for the outside of the phantom image: Average the mean CT value for all specified outside coordinates of one slice.

AvXo

Average outside mean CT number for the number of slices in an exam.

AvSDc

Average image noise about the center image coordinate (measured as the standard deviation) of all slices in an exam.

AvSDo

Average image noise (standard deviation) for the outside of a phantom: Average of all outside coordinates of all the slices in an exam.

6.1.4 Image Series for LightSpeed QX/i System

- 1.) Place the 48cm phantom on the phantom holder.
 - Align phantom with the internal laser lights.
 - Center the phantom with the Calibration program.
- 2.) Manually select the scan parameters in [Table5-9](#) or locate a similar protocol in the Service Protocol - **Image Series 48cm**

Note: This protocol has 16 images (in 4 scans) with an interval of 0.0cm.

Scan Type	Phantom	kV	mA	SFOV	Thickness	Scan Time	Start/End Loc.	# of scans
Axial	48cm	120	340	Large	5mm (4i)	1.0sec	17.5/S7.5	4

Table 5-9 LightSpeed QX/i Image Series Scan Parameters

- 3.) Record the data on the data sheet, and verify the images meet specifications.
- 4.) Proceed to [Section 6.1.5](#), to analyze the image series.

6.1.5 Analyze Image Series

- 1.) Open the Service Desktop.
- 2.) Display the System Test Menu.
- 3.) Select SYSTEM INTEGRATION.
- 4.) Select SYSTEM TESTS.
- 5.) Select IMAGE ANALYSIS.
 - When the Browser opens, select the 48cm Exam/Series/ to analyze.
 - Refer to the Data Sheets (starting on page 57) for the list of images to select (i.e., 1, 5, 9, and 13) before you select the Series means.
 - To select multiple images, press and hold the **CTRL** key while pressing the left mouse button to highlight/select the images.
- 6.) Select SERIES MEANS to open a viewer and results window.
- 7.) The results window reports the series means data. Record the results data on the data sheet, and verify it meets all specifications.

6.1.6 QA Image Series

- 1.) Place the Helios QA phantom, 2144715, on the phantom holder.
 - Align with the black line with the laser lights.
 - Center with the Calibration program.
- 2.) Select the Service Protocol, IMAGE SERIES QA.
-or-
- 3.) Manually select the scan parameters in [Table5-10](#).

Scan Type	Phantom Type	kV	mA	SFOV	Thickness	Scan Time	# of scans	Scan Length	Total imgs
Axial	QA#1	120	130	Small	10mm 2i	2.0sec	4	I5.0 - S5.0	8
Axial	QA#1	120	130	Small	10mm 2i	2.0sec	4	Bone Retro w/ DFOV = 15cm	8
Axial	QA#2	120	260	Small	10mm 2i	1.0sec	4	S400.0 - S50.0 Peristaltic OFF	8
Axial	QA#3	120	260	Small	10mm 2i	1.0sec	4	S55.0 - S65.0	8

Table 5-10 Image Series QA Scan Parameters

- 4.) Retrospectively reconstruct the first QA#1 scan with the Bone algorithm, and 15cm Display FOV, for later analysis. Use the parameters in [Table5-10](#) to reconstruct the image.
- 5.) Record the data on the data sheet, and verify the images meet specifications.
- 6.) Proceed to [Section 6.1.7](#), to analyze the QA image series.
 - a.) Analyze the 4 images acquired with QA#2 at S40.0.
 - b.) Analyze all 8 images acquired with QA#3.

6.1.7 Analyze the QA Image Series

Return to the Image Analysis tool.

- 1.) Select the QA Exam from the Image Browser.
- 2.) Select QA#1 to open a viewer and results window.
- 3.) Refer to [Figure 5-1](#), to adjust the image window:
 - a.) Press and hold the middle mouse button.
 - b.) Move the mouse in the horizontal and vertical directions to adjust the window width and Window level, until the line pair becomes visible.

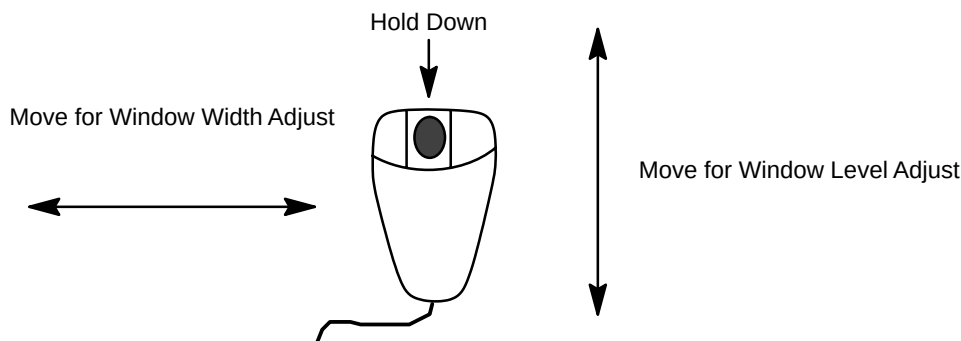
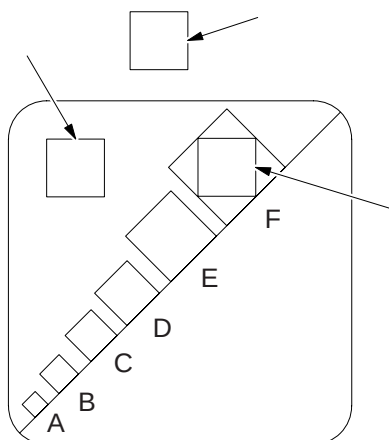


Figure 5-1 Adjust Window Width/Level with Mouse

- 4.) Refer to [Figure 5-2](#). To position the ROI box, and analyze the region of interest:
 - a.) Move: Position the mouse cursor over the box. Press and hold the left mouse button, and move the mouse to reposition the box.
 - b.) Accept: Click the right mouse button once, to accept the current position.
 - c.) Accept: Toggle the right mouse button to accept the remaining ROIs.
- 5.) The system calculates and displays the MTF and Contrast Scale in the Image Analysis Results

window.



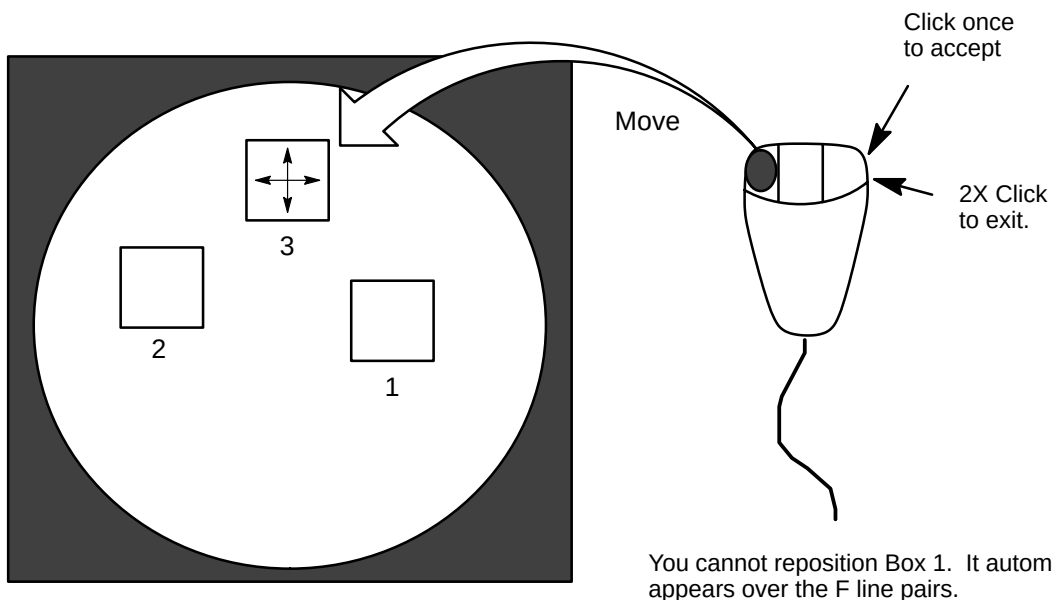
Contrast Scale = mean of B – mean of C.

SD = SD of ROI Box over F slits.

SD = (SD + SC) / 2

Modulation = _____

MTF = 2.2 (Modulation/Contrast Scale)



You cannot reposition Box 1. It automatically appears over the F line pairs.

Figure 5-2 Reposition ROI with Mouse

- 6.) Double click the right mouse button to exit.
- 7.) The software automatically evaluates the QA#2 & 3. Select the Exam/Series and the QA#2 or QA#3 Softkeys to execute.
- 8.) Record the data on the data sheet, and verify the scans meet all specifications.

6.2 Image Data Sheets

6.2.1 Image Series Data Sheet 48cm Poly Phantom

48cm/L Scan parameters: 5mm (4i)/ 340mA /1 sec/16 Images

Exam/Series/ Image	AvXc	AvXo	AvXo - AvXc	AvSDo	Comments	Scan Tech Signature/ Date
___/___/ 1, 5, 9, 13						
2, 6, 10, 14						
3, 7, 11, 15						
4, 8, 12, 16						
SPECS	---	---	±8.5	<40.0	---	---

Table 5-11 48cm Poly Phantom Image Series Data

*If your AvSDo fails, check to make sure you did not scan this technique at 1mm.

Box Size: 45 x 45 pixels

Center Coordinates: 256, 256

Outside Coordinates: 256, 60452, 256256, 45260, 256

Image Acceptance/Date: _____

Certified Image Reviewer: _____

ARTIFACT LIMITS:

Rings: Smudge:

- 48/L; 30 to 36 counts
- 1.25mm
- 48/L; 6.8 counts
- 1.25mm; 12.0 counts

Band: Streaks:

- 8.0 counts
- 4.0 counts

Band Radius: Center Artifact::

- 0 to 23.5cm
- N/A

Clump: Center Artifact::

- 48/L; 3.0 SIGMA
- 1.25mm; N/A
- 1.25mm; 14.0 counts

Center Spot: N/A

6.2.2 Image Series Data Sheet 20cm QA#1 Phantom

QA#1 scan parameters: S/10mm (2i)/120kV/260mA/2 sec/4 images

Use this scan data for Bone Retro.

Record data on **Form 4879**.

Exam/Series/ Image	MTF	MTF 2 slice avg.	Contrast Scale	Comments/ Artifacts	Scan Tech Signature/ Date
___/___/1		MTF1 + MTF3 ÷ 2 =			
3					

Table 5-12 QA#1 Image Series Data

Exam/Series/ Image	MTF	MTF 2 slice avg.	Contrast Scale	Comments/ Artifacts	Scan Tech Signature/ Date
2		MTF2 + MTF4 ÷ 2 =			
4					
SPECS	---	0.65 to 1.0	110.0 to 130.0	---	---

Table 5-12 QA#1 Image Series Data

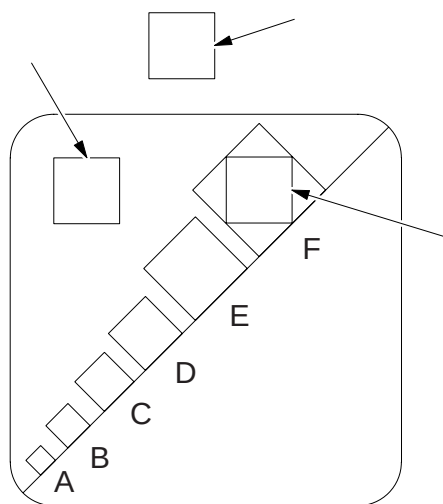
QA#1 scan parameters: S/4x5 (2i)/120KV/130mA/2 sec/4 images

Bone Retro/Display FOV = 15cm

Record data on **Form 4879**

Exam/Series/ Image	Line Patterns Visible	Comments/ Artifacts	Scan Tech Signature/Date
____/____/1			
2			
3			
4			
SPECS	B, C, D, E, F	---	---

Table 5-13 QA#1 Bone Retro Data



Contrast Scale = mean of B–mean of C.

SD = SD of ROI Box over F slits.

SD = (SD + SC) / 2

Modulation = _____

MTF = 2.2 (Modulation/Contrast Scale)

Box Size: 17 x 17 pixels

6.2.3 Image Series Data Sheet 20cm QA#2 Phantom

QA#2 scan parameters: S/4x5 (2i)/120KV/260mA/1 sec/Peristaltic OFF

Scan from S40.0 to S50.0

Record results on **Form 4879**

Exam/Series/ Image at location S40.0	Visible Hole View w/ Window 20	Contrast Factor	Comments/ Artifacts	Scan Tech Signature/Date
____/____/1				
3				
5				
7				
SPECS	See Table 5-15	2.0 to 12.0	---	---

Table 5-14 QA#2 Data Table

Box size: 49 x 9 pixels

Contrast Factor	Visible Number of Holes: Lower Limit*	Visible Number of Holes: Upper Limit*	Smallest Visible Hole: Size
2.00 to 3.99	2	5	7.5mm
4.00 to 7.99	3	5	5.0mm
8.00 to 12.00	4	5	3.0mm

*Required number of visible holes depends on the contrast factor.

Table 5-15 Visible Hole Table

2 out of the 4 scans taken must meet this specification.

Image Acceptance/Date: _____

Certified Image Reviewer: _____

For Reference Only:

Use the manual method for calculating Contrast Factor. only when you cannot access Tool QA#2.

ROI D

ROI E				
10mm	7.5mm	5.0mm	3.0mm	1.5mm

ROI D 49 X 9 pixels x=256, y=200
ROI E 49 X 9 pixels x= 256, y=235
Contrast Factor = Mean E – Mean D

6.2.4 Image Series Data Sheet 20cm QA#3 Phantom

Use Series Means to perform calculations.

QA#3 scan parameters: S/10mm 2i/120KV/260mA/1 sec/8 Images

Scan from S55.0 to S65.0

Record data on **Form 4879**.

Exam/Series/ Image	AVXc 4 scan avg.	Std Dev 4 scan avg.	AvXo - AvXc	Comments/ Artifacts	Scan Tech Signature/Date
____/____/1,3,5,7					
2,4,6,8					
SPECS	0 +3.0	2.6 to 3.4	+3.0	- - -	- - -

Table 5-16 QA#3 Data

Comment: The average value of each row in [Table5-16](#) must meet the specifications listed at the bottom of the corresponding column.

Box Size: 31 x 31 pixels at Coordinate 256, 256

Center Coordinates: 256, 256

Outside Coordinates: 256, 95

417, 257

256, 417

95, 256

ARTIFACT LIMITS:

- **Rings:** 4.0/4.8 cts
- **Band:** 2.8 cts
- **Band Radius:** 0-8.5 cm
- **Clump:** N/A
- **Center Spot:** N/A
- **Center Artifact:** 3.5 Std. Dev.
- **Smudge:** 2.2 counts 1mm; 4.0 counts
- **Streaks:** 4.0 cts

Image Acceptance/Date: _____

Certified Image Reviewer: _____

Chapter 6

Load the Smart Prep Option

Software Options must be loaded every time the OC software is re-loaded. This program must be run once for each option MOD you are loading. For example, if you have two options MODs you must run this program twice.

- 1.) On Service Desktop, select UTILITIES -> INSTALL OPTIONS, and then click START. An Options Window appears.

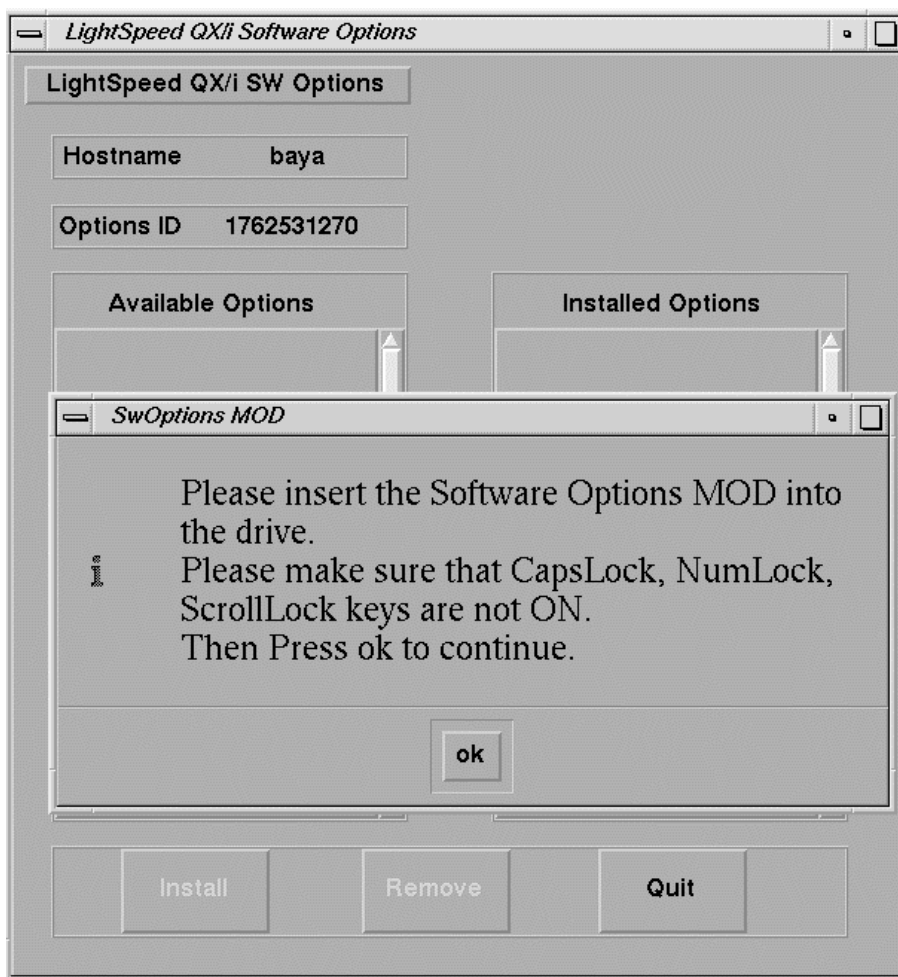


Figure 6-1 Options Window when First Selected

- 2.) Insert the Options MOD into its drive
- 3.) Click on OK

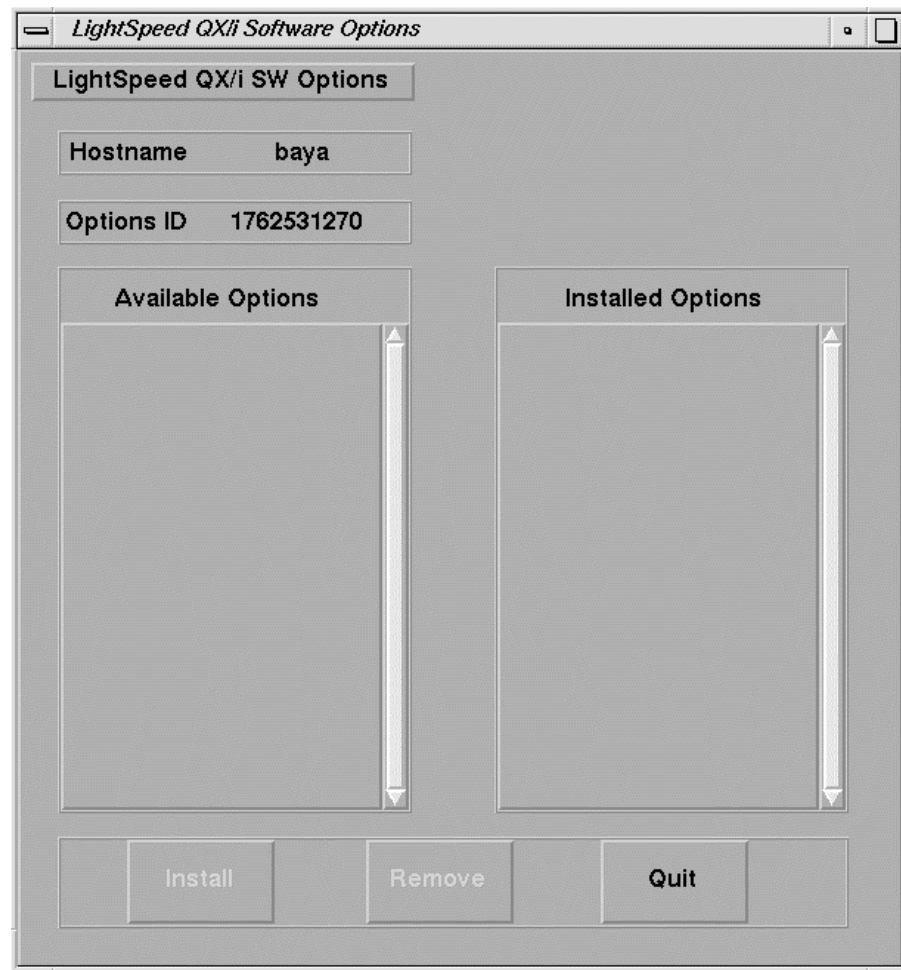


Figure 6-2 Example: Options Window

- 4.) Select the options to be installed. For example, click and drag mouse to highlight all the available options and then release mouse.
- 5.) Click on the INSTALL button. A box may appear while the options are loading. When the option is displayed in the "Installed Options" list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half a minute (due to the fact that they are installing software).
- 6.) After the options are installed, select QUIT, then OK.

If there are no more options to load, a reboot is required to make options active. If you plan to reboot later in the procedure, you may wait until that time.

- 7.) Remove the MOD & write protect the side with options.

The initial install requires the MOD to NOT be write protected; subsequent installs can be done with the MOD write protected.

- 8.) If you have another option MOD to load, go back to [step 1](#).

Chapter 7

Collect DAS Data and Save on MOD

Section 1.0 DAS Data Collection

Time to complete this section: 30 minutes

Section 2.0 Description

CT Engineering has requested that DAS data be collected to get a better understanding of DAS performance and repeatability. The scans are to be collected using Diagnostic Data Collection (DDC) from the Service Desktop. No on-site analysis of the data is necessary, however the Scan Data must be saved on MOD and shipped back to CT Engineering. A MOD has been supplied with this FMI to store the Scan Data.

Section 3.0 Scanning using DDC Protocols

All data to be collected will be from scanning pre-defined protocols from DDC.

- 1.) Bring up the Diagnostic Data Collection (DDC) Main Menu.
- 2.) Go to the Service Desktop.
 - A.) Double Click on TROUBLESHOOT.
 - B.) Double click on DATA ACQUISITION.
 - C.) Double click on DIAGNOSTIC DATA COLLECTION.
- 3.) Verify nothing is within the Scan Beam (Back out the cradle, etc. It is OK to leave the mylar window in place.
- 4.) From the DDC main menu, the list of Protocols are found under the GUI button named Protocol Name
- 5.) Take the following Protocol scans. Refer to next page for DDC GUI Example and list of protocols.

DDC GUI Example

Diagnostic Data Collection

Protocol Name: ddc_static_xray_off

Scan Type

Static X-Ray Off

Static X-Ray On

Rotating X-Ray Off

Rotating X-Ray On

Miscellany

Protocol Name

Position Tube

Show Request

Monitor Enable

Scan Set #0

Prev

Next

Run Description

Scan Time
[0.5 - 4.0] (sec)

No. of Scans

InterScan Delay
(sec)

kV
[80/100/120/130]

mAs
[10 - 400]

DAS Gain
[1 - 31]

Trigger Rate
(Hz)

Filter

Focal Spot

Slice Collimation

Collimation Vector

Options

Auto Scan

1XX1

Gantry Velocity
[1.2/1.4] (sec/rev)

Scan Start Position
(deg/pos)

X-Ray Duration

Qty Unit X-Ray On
[0 - 29.0](sec)

Rotar
RxOverrides

1.00

0.00

On

Off

994

1230

Head

Body

Blocked

Air

Small

Large

4x1.25

4x2.50

4x3.75

4x5.00

None

Save

Accept Rx

Dismiss

Figure 7-1 Diagnostic Data Collection GUI (DDC)

Note: Record Exam numbers and requested information in section Section 6.0 on page 67.

DDC Protocol Scan List

DDC Protocol Name	# Scans	DDC Protocol Name	# Scans
das_dnoise_offsets	2 Scans	das_interconnect3	1Scan
das_popmicro	3 Scans	das_interconnect4	1Scan
das_xrayverif	1 Scan	das_interconnect5	1Scan
das_xrayverif2	1 Scan	das_interconnect6	1Scan
das_xrayverif3	1 Scan	das_interconnect7	1Scan
das_xrayverif4	1 Scan	das_interconnect8	1Scan
das_drift	6 Scans	das_interconnect9	1Scan
das_interconnect	1Scan	das_interconnect10	1Scan
das_interconnect2	1Scan	das_interconnect11	1Scan

Table 7-1 DDC Protocol Scans

Section 5.0 Saving Scan Files to MOD

- 1.) Place the new MOD which are supplied with the FMI, in the MOD Drive.
- 2.) From the Recon Management Menu, select the SAVE scanfiles menu.
- 3.) Select or Highlight only the DDC scans shown on the Data Collection Summary Sheet.

Note: Example of Save Scanfiles Screen (Does not reflect all scans that need to be saved.)

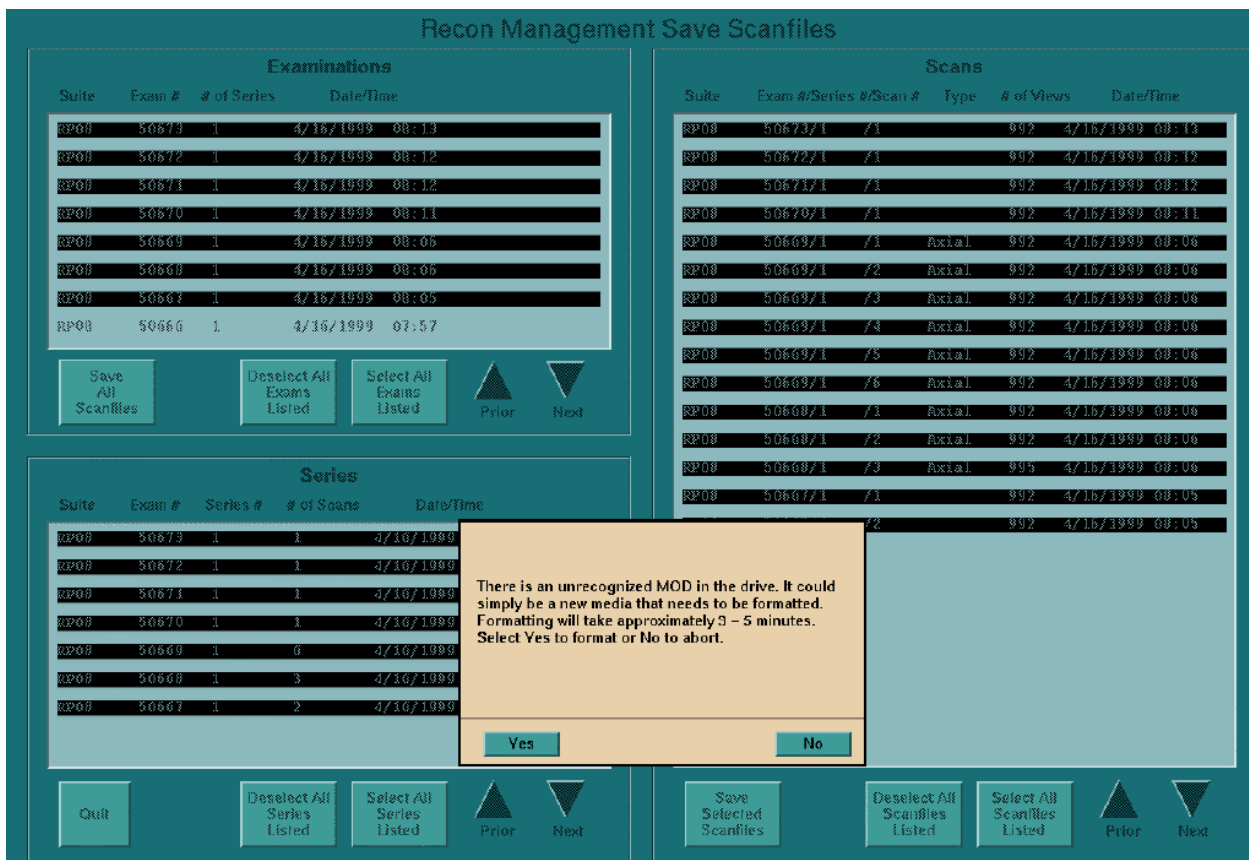


Figure 7-3 Recon Management GUI

- 4.) After successful saving of the data, a message box will be displayed indicating the number of scanfiles saved. The total number of scanfiles to be saved is 26.
- 5.) Label the MOD with Site Name and System ID.
- 6.) Ship the MOD and Data Collection Summary Sheet via Air Courier to:
(Do NOT use regular mail)

Rudy Kopp NB-918 Telephone: (414) 785-5261

GE Medical Systems

16800 W. Ryerson Road

New Berlin, WI. 53151

Section 6.0 Data Collection Tables

Today's Date: _____

Site Name: _____	System ID: _____	Install Date: _____
FE Name: _____	Pager #: _____	E-Mail: _____
Detector S/N: _____	DAS S/N: _____	Gantry S/N: _____

Protocol	Scan Description	Exam #	Series	Scan	Comments
das_dcnoise_offsets	4 x 5.00 / Gain 31		1	1	
***	4 x 1.25 / Gain 5		1	2	
das_popmicro	4 x 5.00 / Gain 31 Scan Speed 1.0 sec.		1	1	
***	4 x 1.25 / Gain 5 Scan Speed 1.0 sec.		1	2	
***	4 x 1.25 / Gain 5 Scan Speed 0.8 sec.		1	3	

Protocol	Scan Description	Exam #	Series	Scan	Comments
das_drift	4 x 5.00 / Gain 31		1	1	
***	4 x 5.00 / Gain 31		1	2	
***	4 x 5.00 / Gain 31		1	3	
***	4 x 1.25 / Gain 5		1	4	
***	4 x 1.25 / Gain 5		1	5	
***	4 x 1.25 / Gain 5		1	6	

Note: The following Protocols utilize X-Ray. Make sure nothing is in the X-Ray beam path.

Protocol	Scan Description	Exam #	Series	Scan	Comments
das_xrayverif	4 x 1.25 / Gain 5		1	1	
das_xrayverif2	4 x 2.50 / Gain 9		1	1	
das_xrayverif3	4 x 3.75 / Gain 29		1	1	
das_xrayverif4	4 x 5.00 / Gain 31		1	1	

Protocol	Scan Description	Exam #	Series	Scan	Comments
das_interconnect	4 x 2.50 / Gain 9		1	1	
das_interconnect2	4 x 3.75 / Gain 29		1	1	
das_interconnect3	4 x 5.00 / Gain 31		1	1	
das_interconnect4	4 x 1.25 / Gain 5		1	1	
das_interconnect5	Cal 1 / Gain 5		1	1	
das_interconnect6	Cal 2 / Gain 5		1	1	
das_interconnect7	Cal 3 / Gain 5		1	1	
das_interconnect8	Cal 4 / Gain 5		1	1	
das_interconnect9	Cal 5 / Gain 5		1	1	
das_interconnect10	Cal 6 / Gain 5		1	1	
das_interconnect11	Cal 7 / Gain 5		1	1	

Ship the MOD and Data Collection Summary Sheet via Air Courier to:

(Do NOT use regular mail)

Rudy Kopp NB-918 Telephone: (414) 785-5261

GE Medical Systems

16800 W. Ryerson Road

New Berlin, WI. 53151

Section 7.0 Disposal of old Materials/Parts

The material left over from the FMI should be disposed of using normal service protocols. Please return all necessary recyclable parts to the Renewable Resources team in your respective areas. Many of the parts are still useful for repair of equipment and are not obsolete.

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Chapter 8

Save the System State on both the local Hospital MOD.

Before leaving, you will want to save the system configuration information, characterization, calibration, protocols, etc.

- 1.) Bring the system up if it is not already up.
- 2.) Click on the Service desktop.
- 3.) Click on PM.
- 4.) Highlight SYSTEM STATE.
- 5.) Click on the START button.
- 6.) Click on ALL. This will highlight Cals, Characterization, Reconfig Info, etc.
- 7.) Click on SAVE.

The save will take a few minutes. Review the output for errors or missing files; the scroll bar on the right works only when the tool isn't busy performing some task, it may take a little while. If you see any missing files or failures, then refer to the notice below.

- 8.) Click on DISMISS.

If System State reported no problems, then you may proceed to next chapter.

IF YOU SHOULD GET SAVE/RESTORE SYSTEM STATE ERRORS:

The Save/Restore System State process reports status on each file specification listed in the /usr/g/config/SysState.cfg file. The status may be:

- Save (or Restore) of {filename} succeeded
- Save (or Restore) of {filename} skipped. File not found
- FAILED

A status of FAILED, indicates potential problems, skipped means the file not found was not required. Please review the following for each file you see an error reported.

Some files simply may not exist on the system. If certain tools were never run then certain files may never get created. You may view /usr/g/config/SysState.cfg and look for a required field. req=Y means the file in question is required. You may also wish to check the system and MOD to verify whether it was actually missing or if there were permission or owner problems with the file.

SOME COMMANDS YOU MIGHT FIND HANDY:

<code>mountMOD</code>	<i>(mount the MOD on the /MOD dir)</i>
<code>ls -l /MOD/service_mod_data/system_state</code>	<i>(or /MOD/serv*/sys*)</i>
<code>find /MOD -name '*abc*' -print</code>	<i>(locates any file with 'abc' in it)</i>
<code>cd; unmountMOD</code>	<i>(must cd out of MOD dirs to unmount it)</i>
<code>more /usr/g/config/SysState.cfg</code>	<i>(view locations of System State files)</i>
<code>su -; (password); man 1 chmod chown</code>	<i>(access man pages as root)</i>
<code>rsh sbc</code>	<i>(access the SBC)</i>

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Chapter 9

DAS Cable Modifications

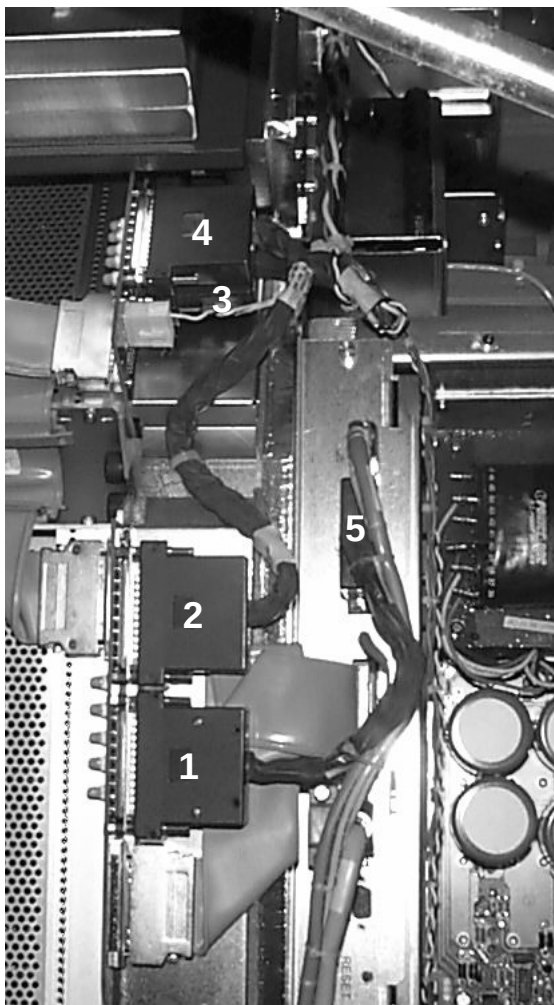
Section 1.0 Introduction

The LightSpeed DAS power cables currently have a problem where a restraining bar inside the connector loosens and causes intermittent shorts. The electrical shorts cause intermittent problems that may result in random resets of the DCB or other boards powered by the DAS power supplies.

The following instructions show how to identify the cables, disassemble the covers, and remove the metal bar inside the connector.

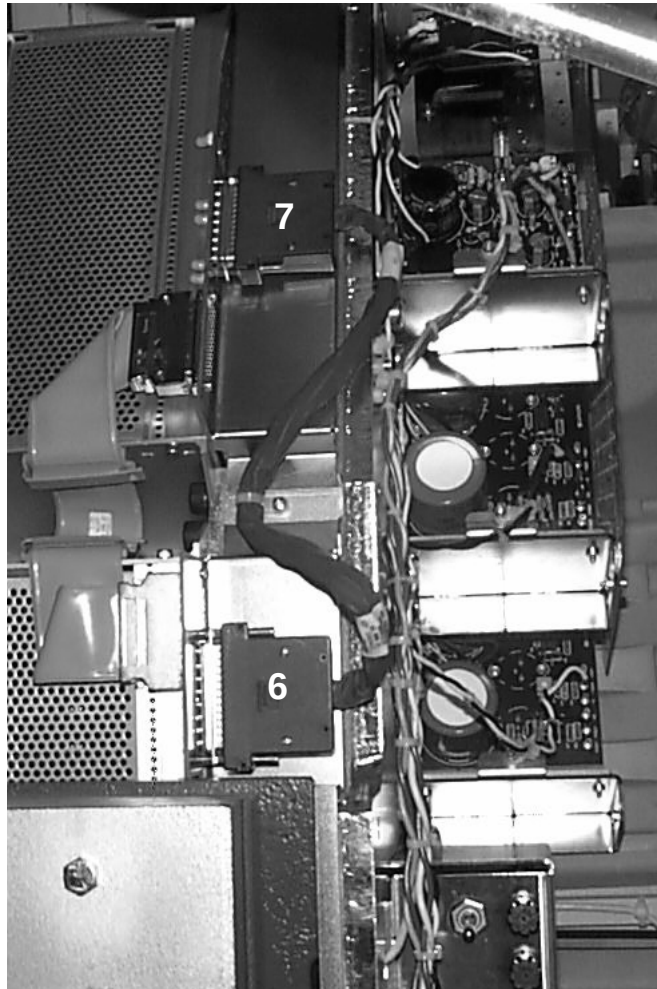
Section 2.0 Locating the Power Cables

- 1.) Following proper lock and tag out procedures, open the gantry and position the DAS at the service side of the gantry.
- 2.) Locate the cables identified in [Figure 9-1](#) and [Figure 9-2](#). There are several connectors on the cables that provide power to the DAS.



There are several connectors identified in the illustration. They are labeled 1, 2, 3, 4, and 5. Connector 3 is tied into a larger harness that can not be removed.

Figure 9-1 DAS Cable Locations



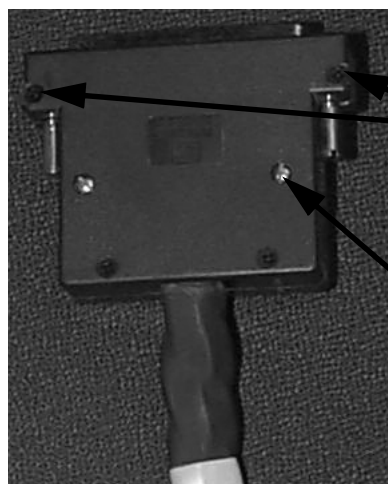
There is a single cable shown in this illustration. Connectors are labeled 6 and 7.

Figure 9-2 DAS Cable Locations

- 3.) Remove the cables from the gantry. It is easier to access the cables outside the gantry. One cable will not be able to be removed since it is tied to a larger harness.

Section 3.0 Disassembly of connector

- 1.) Remove the four Philip head screws as identified in [Figure 9-3](#).

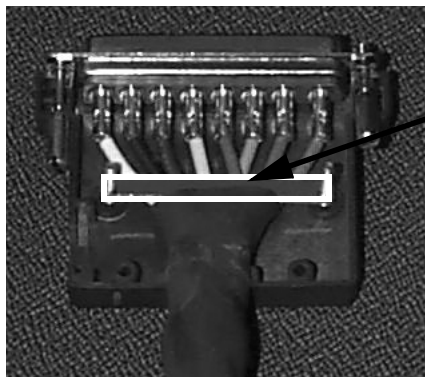


Four Philip Screw holding Connector casing together.

Two flat blade screws holding restraining bar together.

Figure 9-3 Connector Assembly

- 2.) Pull off shell. Locate the restraining bar inside the connector. Refer to [Figure 9-4](#).



Restraining bar inside connector to be removed.

Figure 9-4 Restraining Bar in Connector

- 3.) Remove the restraining bar by removing the flat blade screws as seen in [Figure 9-3](#). Dispose of the two screws and the metal restraining bar. They will no longer be assembled inside the connector.

Section 4.0 Reassemble connector and install cables

- 1.) Reassemble the connectors with the four Phillip screws.
- 2.) Install cables in gantry per [Figure 9-1](#) and [Figure 9-2](#).
- 3.) Reassemble gantry covers and take a few scans to verify system operation.

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Chapter 10

Detector and DAS Handling - ESD Considerations

Section 1.0 Introduction

The LightSpeed (QX/I) CT scanner is a premium-tier, 3rd generation CT scanner. It's distinguishing feature is the ability to simultaneously collect 4 rows of scan data. This 4-row data collection is accomplished via a 16-row detector and a 4-row DAS. This detector/DAS combination will provide functionality unmatched by any CT scanner in the world. This new, multi-slice technology does not come without compromise. In order to support the Detector row selection, and physically support the number of detector channels, the detector assembly requires LSI components and compressed signal traces. As a result, the detector is extremely susceptible to physical handling techniques.

Section 2.0 Electrostatic Discharge

The LightSpeed Detector Assembly has 57 Modules. Each Module contains electronic components which are electrically sensitive. An electrostatic discharge (ESD) between 100 to 1000 V may damage a component. This is substantially less than the 3000 V discharge needed to feel any static. The ESD may cause an immediate failure, or it may weaken components to produce future, intermittent problems. To avoid or substantially reduce ESD damage to the detector, which may result in detector replacement, always adhere to the proper LightSpeed Detector ESD guidelines when servicing the Detector or Detector Flex to DAS Backplane interface:

- Accumulate all necessary tools and ESD items before servicing the System. Keep them within reach to avoid Ground strap removal, stirring up dust, creating air friction, etc....
- Always verify DAS power is turned off before servicing the DAS, Detector, or their Flex interface.
- Always wear a ground wrist strap grounded to the Gantry. The use of a Ground Monitor (3M-725) will confirm a good ground connection.
- Wear suggested finger cots to isolate ESD and dirt/oil from fingers.
- Never leave a flex "free-floating". Cover Flex Contact end with ESD "boots" after disconnecting from Backplane housing.
- Only use recommended Alcohol wipes and clean compress air when cleaning debris from flexes, housings, elastomers, and covers. Never use other chemical additive cleaning solutions.
- Always torque Flex housing cover clamps to 9 in-lbs.
- It's essential to have a good clean contact between the Detector Flex and DAS Backplane to reduce noise during DC Noise tests and micro phonics, as well as preventing opens or shorts.

Section 3.0 Physical Handling

Since the LightSpeed Detector has 57 Detector modules within it's assembly, and each module has 2 Flex leads extending from the detector, the detector has to be handled with extreme care.

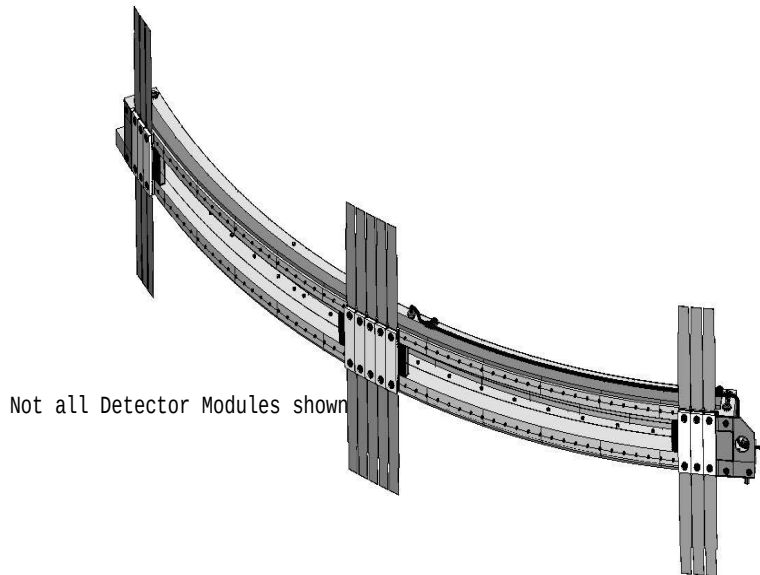


Figure 10-1 LightSpeed Detector Illustration

The Detector can only be safely located either mounted in the Gantry or in the replacement Detector shipping container. Due to the flexes extending from it's sides, the detector cannot be set down on a table or other surface. Damage to the flexes can occur.

Section 4.0 Detector Handling Guidelines

- Always wear a ground wrist strap grounded to the Gantry, the Detector, and to the replacement Shipping container when installing or removing a detector. The use of a Ground Monitor (3M-725) will confirm a good ground connection.
- Use care in removing/ installing the Air Plenum. The ends of the Air Plenum can interfere or "snag" on the detector flexes or with the Thermal protection switch on the low channel end of the Detector.
- WATCH the flexes when installing or removing the detector. The "free" flexes must have ESD boots installed and extra care must be taken so the flexes don't hit or "catch" on anything during movement of the detector.

- The top covers which hold the flexes to the housings have alignment pins on the top side. These are used to position the clamp in place. Never install the top covers up-side down. The alignment pins **WILL** damage the flex, which if severe enough will require a Detector replacement.

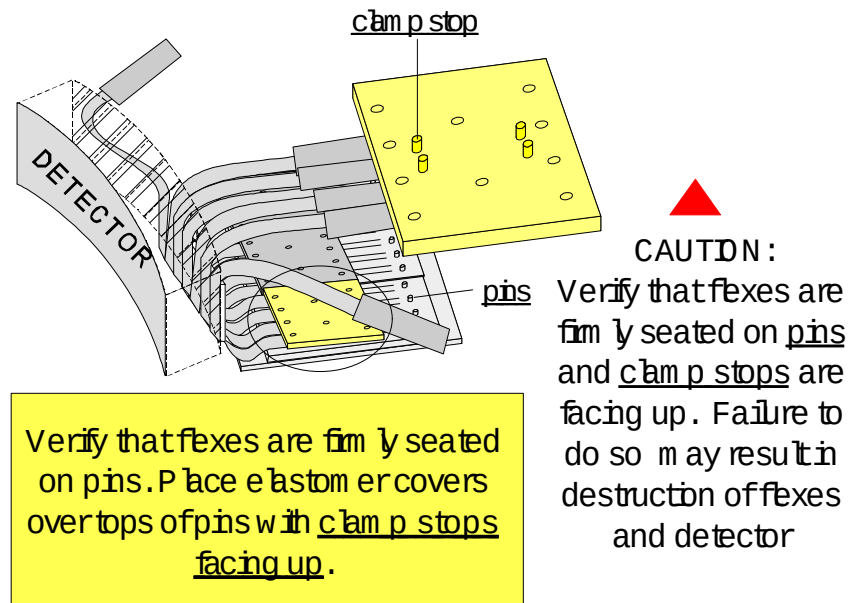


Figure 10-2 Clamp Stops

- Always use the aid of the hoist to lift the detector into/from the Gantry. Remember to disconnect the Detector heater cable (J3) before removing the detector.
- Flexes should never be twisted or bent other than their normal installed position. Never try to bend a flex to try to get it to reach to the adjacent backplane position during troubleshooting. Doing so will damage the flex resulting in detector replacement.
- Never bend the end of the flex near it's contact end. Bending the flex at the Kapton and copper transition area can cause the copper to crack resulting in open detector signals for that flex, thus requiring a detector replacement.

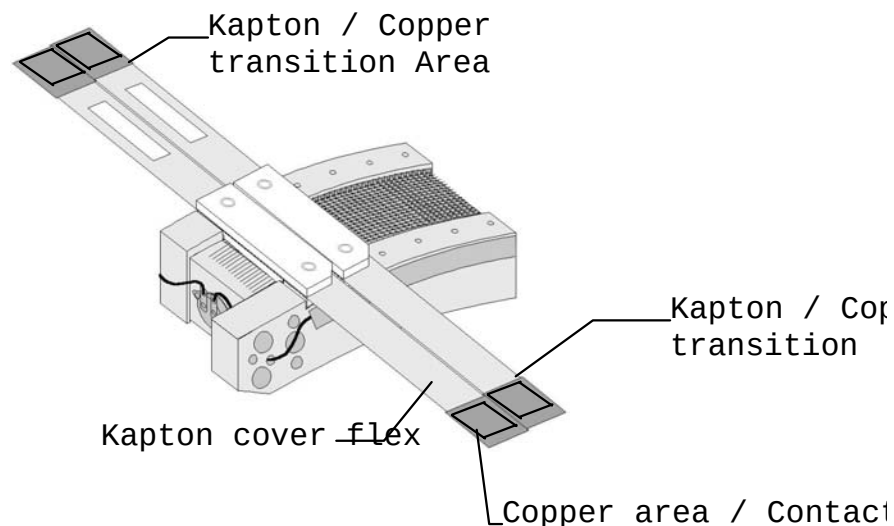


Figure 10-3 Detector Flex

Section 5.0 DAS/Detector Tools for Troubleshooting

DAS/DETECTOR INTERFACE KIT (2227089)

ITEM	PART NUMBER	QTY	COMMENTS
Alcohol Pads	46-183039P1	1	1 Box
Elastomers	2169110-2	20	Enough for 2 housings
Finger Cots (Medium)	2221235	10	
Finger Cots (Large)	2221235-2	10	
Detector Flex Boots	2185881	8	Enough for 2 Housings

Table 10-1 DAS/Detector Kits

DETECTOR ESD KIT

ITEM	PART NUMBER	QTY	COMMENTS
Continuos workstation ESD monitor	2218386	1	3M-725
Fabric Wrist Band, adjustable, dual conductor for monitor	2218387	1	3M-2368
Wrist Strap grounding cord, dual conductor - 5ft	2218388	1	3M-2360
Wrist Strap grounding cord, dual conductor - 10ft	2218390	1	3M-2370
Belt Clip for 725 monitor - 10 clips /pack	2218391	1	3M-723
Complete Kit	2220482	1	3M-8507

Table 10-2 ESD Kit for LightSpeed DAS/Detector Troubleshooting

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